Certifying Prediction Limits and Representation Refinement with Kernel Affine Hull Machines

Mohit Kumar MOHIT.KUMAR@UNI-ROSTOCK.DE
University of Rostock, Rostock, Germany
Software Competence Center Hagenberg, Hagenberg, Austria

Somayeh Kargaran SOMAYEH.KARGARAN@SCCH.AT
Software Competence Center Hagenberg, Hagenberg, Austria

Abstract

We study whether a fixed representation can support a prescribed forecasting accuracy and whether a more costly representation is necessary. Our approach combines Kernel Affine Hull Machine (KAHM) features with decoders whose sensitivity is bounded. A pairwise lower bound detects target separations that exceed the permitted separation between predictions. We characterize when this bound is exact and construct distributions for which it is arbitrarily loose. Pairing within predefined strata can improve the bound while preserving the population risk being evaluated. Finite-sample certificates account for estimated stratum probabilities and combine with upper risk bounds to accept fitted forecasts, exclude decoder classes, or certify minimum feasible representation cost. Appending feature blocks under a sum-of-block-norms budget preserves earlier predictors without enlarging the class when a block is duplicated. Experiments compare the pairwise certificate with variance-based bounds and bounds obtained by minimizing empirical risk; the latter are often tighter. A controlled forecasting experiment demonstrates cost and infeasibility decisions, although substantial calibration samples are required. Applications include six synthetic systems, pen trajectories, and speech energy. KAHM performance depends on the task and decoder. All guarantees concern expected squared error under an explicit sampling law.

Keywords: Kernel Affine Hull Machines, time-series forecasting, soft state abstraction, representation refinement, prediction risk certification

1 Introduction

A forecasting representation maps an observed history to features, and a decoder maps those features to future observations. A representation can reconstruct past observations accurately while discarding distinctions needed to predict their futures. This raises two questions: does a fitted forecast meet an error tolerance, and can any decoder with a specified sensitivity meet it using that representation? We address both for expected squared error on a fixed observed signal. We call a statistically valid bound used to answer one of these questions a *risk certificate*. Comparisons between representations use the same target and evaluation distribution.

Kernel Affine Hull Machines (KAHMs) provide our principal application. Their local reconstruction models produce soft associations between a history and learned reference groups (Kumar et al., 2024, 2025). A decoder maps these associations to future observations. We retain the reconstruction mechanism of Kumar and Kargaran (2026), who analyze prediction of successor soft coordinates and independent-trajectory evaluation. We instead study observed outputs, general sensitivity-constrained decoders, and the cost of refining a representation. The analysis does not require the feature vector

©2026 Mohit Kumar and Somayeh Kargaran.

License: CC-BY 4.0, see https://creativecommons.org/licenses/by/4.0/.

to evolve through an exact linear transition, to be a Markov state, or to preserve all predictive information in the history.

The lower certificate starts from a simple obstruction. If two targets are farther apart than the decoder's Lipschitz constraint allows their predictions to be, their prediction errors cannot both be small. We call this discrepancy a *pairwise obstruction*: it is the excess target separation beyond what the sensitivity budget permits. Averaging its squared positive part over conditionally independent pairs gives a population lower bound, which we call a *witness*. Related pairwise Lipschitz regression arguments are established (Akash et al., 2021); our analysis identifies the resulting witness's slack, its dependence on the pairing law, and finite-sample decisions under a fixed marginal risk. The witness is sharp for some distributions and can be arbitrarily loose for others. This limitation motivates stratified pairing: concentrate comparisons within predefined groups, then reweight by their population masses. Estimating these masses adds uncertainty that must enter the certificate.

Upper and lower certificates have complementary decision value. An upper bound can certify a fitted model's accuracy. Lower bounds can exclude cheaper representation classes, making it possible to certify the minimum feasible cost among predefined alternatives, or their joint infeasibility. This is a stronger conclusion than selecting the cheapest fitted model whose upper bound passes. It does not require the selected predictor itself to change.

Our contributions are as follows.

1. **Witness analysis and stratified certification.** We give an exact slack decomposition, sharp continuous and discrete cases, and counterexamples to universal tightness. A stratified certificate accounts for random stratum counts and estimated masses while retaining the original population risk.

2. **Refinement and decision guarantees.** Retaining feature blocks under a sum-of-block-norms budget preserves earlier predictors and makes exact duplication immaterial to the feasible class. Simultaneous lower and upper bounds yield accuracy, improvement, and minimum-cost statements.

3. **Controlled comparisons and forecasting evidence.** A fixed benchmark compares the witness with a range-aware centered bound and a certified empirical-minimum-risk bound for the same Lipschitz class. A separate experiment isolates decisions enabled by lower calibration. KAHM applications cover six synthetic systems, original pen trajectories, and speech energy, with matched geometric controls and conventional regression baselines.

The empirical findings delimit the claims. Empirical-minimum calibration is often tighter than the pairwise certificate, and stratification can improve or weaken a lower bound. KAHM accuracy gains depend on the task and decoder. Informative real-data certification requires many defensible independent groups and a stated sampling assumption. The guarantees concern expected risk under that evaluation law, not prediction-interval coverage for every individual history.

2 Related work

Geometric representations and Koopman prediction. KAHMs learn reconstruction models in reproducing kernel Hilbert spaces (RKHSs) (Kumar et al., 2024, 2025). We use these established reconstruction models to construct forecasting features. Soft state aggregation (Singh et al., 1994), approximate state abstraction (Abel et al., 2016), and task-oriented latent transition models (Gelada

et al., 2019) likewise distinguish a representation from its predictive use. Extended dynamic mode decomposition approximates Koopman operators on finite dictionaries (Williams et al., 2015); learned dictionaries (Li et al., 2017) and RKHS regression bounds (Kostic et al., 2022) provide related forecasting frameworks. Basis-invariant consistency diagnostics (Haseli and Cortés, 2023), multistep projection-error guarantees and refinement (Conradie et al., 2026), and residual-based dictionary learning (Coote and Colbrook, 2026) address dictionary quality. Our decisions instead compare expected observed-output error with a declared tolerance while preserving a common target and sampling law.

Closest KAHM prediction-limit analysis. Kumar and Kargaran (2026) study norm-bounded linear prediction of successor soft coordinates, centered lower certificates, empirical-risk-based alternatives, fitted-risk upper bounds, and independent-trajectory evaluation. We use their KAHM reconstruction and sampling principle. Our analysis concerns observed-output error and a positive-part pair witness. It characterizes the witness's slack, studies stratification, and derives cost certificates for classes that preserve earlier feature blocks. The class is unchanged when an existing block is duplicated. Independent-episode calibration and acceptance based on an upper risk bound are established components of this framework. Our experiments compare both centered lower bounds and lower bounds obtained by certified empirical-risk minimization.

Pairwise bounds and Lipschitz regression. Akash et al. (2021) derive a Lipschitz-constrained mean squared error (MSE) bound using paired residuals and target separation. Their argument is a close antecedent to the geometric step in our witness; our contribution concerns its conditional positive-part form, sharpness and slack, and statistically valid stratification under the unchanged marginal risk. Finite-support Lipschitz regression can be posed as a constrained optimization problem (Gottlieb et al., 2017); we use this class directly for an empirical-minimum comparator. Model-class falsification (Müller et al., 2025) and minimal-empirical-risk bounds (Wu and Yang, 2026) provide broader context. The Gamma test uses local pair differences to estimate regression noise (Evans and Jones, 2002). Lipschitz set-membership identification (Milanese and Novara, 2004), trajectory invalidation (Jin et al., 2020), and corruption-robust prediction ranges (Blum and Nguyen, 2026) concern related feasibility questions with different sampling and error criteria.

Calibration, refinement, and dependence. We use classical bounded-mean inequalities (Hoeffding, 1963) and empirical Bernstein and standard-deviation bounds (Maurer and Pontil, 2009). The simultaneous testing principle follows Learn then Test (Angelopoulos et al., 2025). Our group norm is established in group selection and multiple-kernel learning (Bach, 2008); here its max-block dual metric gives a comparable sensitivity budget as dictionaries grow. Soft gating and residual fitting also have established precedents (Jordan and Jacobs, 1994; Bühlmann and Yu, 2003). The additional cost conclusion comes from jointly excluding cheaper classes and accepting a fitted model.

Within-episode dependence is allowed because concentration is applied to independent episode summaries. A chronological split of one dependent record does not provide that independence. Forecasting bounds under other sampling assumptions (McDonald et al., 2017) and dependence-aware conformal analysis (Oliveira et al., 2024) illustrate the additional conditions needed; their guarantees are not consequences of the present theorem.

3 Learning and certifying KAHM forecasts

The procedure has three stages. KAHM reconstruction models map each history to geometric features; a constrained decoder maps those features to future observations; and independent calibration assesses the resulting forecasts. Keeping these stages separate also allows alternative representations to use the same decoder and evaluation procedure.

3.1 Notation and basic conventions

Vectors are columns unless they appear as rows of a data matrix; $v^\top$ denotes transpose. For a vector v, $\|v\|_2$ is its Euclidean norm and $\|v\|_1 = \sum_\ell |v_\ell|$. For a matrix M, $\|M\|_{\mathrm{F}} = (\sum_{k,\ell} M_{k\ell}^2)^{1/2}$ is its Frobenius norm and $\|M\|_{\mathrm{op}} = \sup_{\|v\|_2=1} \|Mv\|_2$ is its operator norm. We use $\langle u, v\rangle = u^\top v$ and $\langle M, N\rangle_{\mathrm{F}} = \mathrm{tr}(M^\top N)$ for vector and matrix inner products; tr sums diagonal entries. The symbols $\mathbf{1}_N$, I_N, and e_ℓ denote the all-ones vector, identity matrix, and ℓth coordinate vector, respectively, with the ambient dimension specified by context.

For a real number z, $z_+ = \max\{z, 0\}$ is its positive part. Logarithms are natural, and $\lfloor z \rfloor$ is the largest integer not exceeding z. For a finite set S, $|S|$ denotes its cardinality; for an event E, $\mathbf{1}_E$ is one on E and zero otherwise. We write $\mathbb{E}$ and $\Pr$ for expectation and probability under the specified sampling law, and $\mathrm{Var}(V) = \mathbb{E}(V - \mathbb{E}V)^2$ for the variance of a real random variable V. For a square-integrable random scalar or vector V, $\|V\|_{L^2} = (\mathbb{E}\|V\|_2^2)^{1/2}$; for scalars, the norm inside the expectation is absolute value. An overbar denotes a sample average, and a hat denotes a fitted or estimated quantity. The projection $\Pi_{\mathcal{Y}}(v)$ is the closest point to v in a nonempty closed convex set $\mathcal{Y}$ in Euclidean distance. For a box, this operation clips each coordinate to its allowed interval. All random variables and prediction maps are measurable.

Table 1 collects the main quantities. The forecasting construction and the statistical statements below give their precise domains and sampling assumptions.

3.2 A fixed raw-output forecasting task

Let $y_t \in \mathbb{R}^{d_y}$ be the observation at integer time t, where $d_y \geq 1$ is the number of observed coordinates. For a history length $m \geq 1$, define the column vector

$$x_t = (y_{t-m+1}^\top, \ldots, y_t^\top)^\top \in \mathbb{R}^{md_y}.$$

Covariates available at time t may be appended to this vector. A forecast origin is a time t at which a prediction is made using only this available information. Let $\mathcal{H}$ be a finite, nonempty set of positive integer horizons, ordered increasingly, and let $H = |\mathcal{H}|$ be its size. The joint target is

$$Y_t = (y_{t+h}^\top : h \in \mathcal{H})^\top \in \mathbb{R}^q, \qquad q = d_y H.$$

Thus each training row contains all requested future observations in the same order. The theory also considers one horizon at a time, in which case its target dimension is d_y (or one for a separately scored coordinate).

The history length, any fixed target transformation, horizons, and origin sampling rule define the prediction task. Here "raw output" means the observed signal under that fixed transformation, rather than learned feature coordinates. When comparing history lengths, we use the same origins, starting after the longest history is available. Every candidate is therefore evaluated under the same target and origin law.

Symbol	Meaning
y_t, d_y	Observation at time t and its dimension.
x_t, m	Causal history and number of observed time steps in it.
$\mathcal{H}$, H, q	Set of forecast horizons, its size, and target dimension; for a joint forecast, $q = d_y H$.
Φ_b, C_b	Feature block b and its number of coordinates.
Γ_s, p_s	Stack of the first s feature blocks and total dimension $p_s = \sum_{b=1}^{s} C_b$.
a, W_b	Decoder intercept in $\mathbb{R}^q$ and block coefficients in $\mathbb{R}^{C_b \times q}$; predictions use $W_b^\top \Phi_b$.
κ, λ	Decoder sensitivity budget and ridge regularization parameter.
$R(f)$, R^*	Population mean squared error of a fitted forecast and infimum over a stated decoder class.
$W_{\kappa,A}$	Population pairwise obstruction, using budget κ and conditioning variable A; distinct from coefficient matrices W_b.
n, K	Number of calibration episodes and number $K = \lfloor n/2 \rfloor$ of disjoint episode pairs.
$Z_{i,jh}$, $Q_{r,jh}$	Loss of candidate j at horizon h on episode i, and obstruction score on pair r.
L_{jh}, U_{jh}	Lower obstruction endpoint and upper fitted-risk endpoint; the unindexed $U = \Gamma(X)$ instead denotes a random representation.
D_h, B_h	Diameter of the known horizon-h output set and squared-loss bound $B_h = D_h^2$.
$c(X)$, π_a, N_a, K_a	Frozen stratum label, population stratum mass, sample count, and within-stratum pair count.
W_{str}, L_{str}, $\mathcal{P}$	Stratified population witness, its lower endpoint, and the confidence set for stratum masses.
c_j, c_-, c_+, c^*	Declared class cost, lower and upper certified cost endpoints, and minimum feasible cost.
ϵ_h, δ, α	Requested RMSE tolerance, total failure probability, and failure probability allocated to one confidence statement.

Table 1: Recurring notation. Mean squared error (MSE) is an expected squared Euclidean error; root-mean-square error (RMSE) is its square root. Experiment-specific coordinate averaging is stated separately.

The main learner directly predicts each requested horizon, sharing a geometric representation and a joint coefficient budget. It does not generate an entire trajectory by iterating one-step forecasts. The rollout baselines instead iterate a fitted feature-transition model and decode the resulting features. Both approaches are evaluated on the same observed targets; a small one-step feature error alone is not assumed to guarantee an accurate long rollout.

An *episode* is one complete trajectory or other prespecified independent sampling unit. Development data comprise training data, used to fit features and coefficients, and validation data, used to choose hyperparameters or a representative within each model family. Calibration data provide the confidence bounds and may select among candidates frozen before calibration. A separate test set

describes forecasting performance after these decisions. Its empirical error estimates population risk; it is not an exact risk value.

3.3 KAHM dictionaries and stable soft associations

A dictionary block describes a history through its associations with a set of reference groups. KAHM associations depend on local reconstruction models. Radial basis function (RBF) associations instead depend on distances to group centers; affine-hull associations use residuals from fitted local affine subspaces together with a locality term. Each construction specifies which histories receive similar coordinates. A soft association retains a weight for every group, while a hard assignment retains only the selected group. These geometric choices need not preserve all information relevant to forecasting.

For block b, partition the training histories into $C_b \geq 1$ reference groups, indexed by $c = 1, \ldots, C_b$. Let $\mathcal{X}$ be the history domain and let $z(x) \in \mathbb{R}^{d_x}$ be the preprocessed history, with dimension d_x. Each group has $J_{bc} \geq 1$ local KAHM reconstruction models, fitted using its training histories (Kumar et al., 2024, 2025; Kumar and Kargaran, 2026).

For one such local model, suppress the indices (b, c, j) and write its $N_{\mathrm{loc}} \geq 2$ reference histories as $z_1, \ldots, z_{N_{\mathrm{loc}}} \in \mathbb{R}^{d_x}$. The fitted linear encoding $P \in \mathbb{R}^{r \times d_x}$ maps histories to $r \geq 1$ principal-component coordinates. The encoding retains only principal-component directions with positive empirical variance. Let $\Sigma \in \mathbb{R}^{r \times r}$ be the resulting nonsingular sample covariance of the encoded references. The Gaussian kernel, its reference Gram matrix, and its query vector are

$$k(\xi, \xi') = \exp\left\{-\frac{(\xi - \xi')^\top \Sigma^{-1} (\xi - \xi')}{2r}\right\},$$
$$K_{i\ell} = k(Pz_i, Pz_\ell), \qquad k_x = (k(Pz_i, Pz(x)))_{i=1}^{N_{\mathrm{loc}}}.$$

Here $\xi, \xi' \in \mathbb{R}^r$. For the local regularization parameter $\eta > 0$, kernel regression gives $h(x) = (K + \eta I_{N_{\mathrm{loc}}})^{-1} k_x \in \mathbb{R}^{N_{\mathrm{loc}}}$. Its coordinate h_i uses regression targets equal to one at reference i and zero at the other references. The reconstruction and its normalizer are

$$s(x) = \sum_{i=1}^{N_{\mathrm{loc}}} h_i(x), \qquad \widetilde{s}(x) = \begin{cases} s(x), & s(x) \neq 0, \\ 1, & s(x) = 0, \end{cases}$$
$$\widehat{z}(x) = \sum_{i=1}^{N_{\mathrm{loc}}} \frac{h_i(x)}{\widetilde{s}(x)} z_i.$$

The training algorithm determines the reference subset, P, Σ, and η; η is separate from the forecast-decoder ridge parameter below. When $s(x) \neq 0$, the reconstruction coefficients sum to one and may be signed, so $\widehat{z}(x)$ is an affine combination of reference histories. The zero-sum convention defines the implementation at all queries but need not preserve this affine-combination property.

Restoring the indices, let $\widehat{z}_{bcj}(x)$ denote the reconstruction from local model j of group c in block b. Its folding score combines reconstruction error and angular discrepancy (Kumar and Kargaran, 2026, Section 3.3). Define

$$e_{bcj}(x) = 1 - \exp\{-\|z(x) - \widehat{z}_{bcj}(x)\|_2\}, \qquad \theta_{bcj}(x) = \frac{\arccos \gamma_{bcj}(x)}{\pi},$$

where $\gamma_{bcj}(x)$ is the cosine similarity $\langle z(x), \widehat{z}_{bcj}(x) \rangle / (\|z(x)\|_2 \|\widehat{z}_{bcj}(x)\|_2)$, clipped to $[-1, 1]$; when either vector has zero norm, the implementation sets this similarity to zero. The inverse cosine $\arccos$

takes values in $[0, \pi]$. The group score is

$$T_{bc}(x) = \min_{1 \le j \le J_{bc}} \left\{ \frac{e_{bcj}(x)^2 + \theta_{bcj}(x)^2}{2} \right\}^{1/2}.$$

Thus $T_{bc} : \mathcal{X} \to [0, 1]$ is a geometric discrepancy score: smaller values indicate a closer reconstruction match. It is not a class probability. Taking the minimum allows any local model within the group to provide the match.

Fix an exponent $\omega > 0$ and offset $\tau > 0$. The exponent controls the contrast between affinities, and the offset keeps them positive. The reported runs use $\tau = 10^{-6}$. Define

$$\Phi_{bc}(x) = \frac{(1 - T_{bc}(x) + \tau)^{\omega}}{\sum_{r=1}^{C_b} (1 - T_{br}(x) + \tau)^{\omega}}, \qquad \Phi_b(x) = (\Phi_{b1}(x), \ldots, \Phi_{bC_b}(x))^{\top}. \tag{1}$$

Then $\Phi_b(x)$ lies in the probability simplex

$$\Delta_{C_b} = \{v \in \mathbb{R}^{C_b} : v_c \ge 0 \text{ for all } c,\ \mathbf{1}_{C_b}^{\top} v = 1\}.$$

The coordinates express relative reconstruction affinity and need not be calibrated probabilities. To evaluate (1) stably, set $\ell_{bc}(x) = \omega \log(1 - T_{bc}(x) + \tau)$ and $\ell_b^{\max}(x) = \max_c \ell_{bc}(x)$, then compute

$$\Phi_{bc}(x) = \frac{\exp(\ell_{bc}(x) - \ell_b^{\max}(x))}{\sum_{r=1}^{C_b} \exp(\ell_{br}(x) - \ell_b^{\max}(x))}.$$

Subtracting the maximum leaves the normalized weights unchanged and makes at least one exponent equal to zero. Consequently, the denominator is at least one in exact arithmetic, even when all unnormalized affinities are small. This preserves relative affinities without a small-denominator replacement by uniform weights.

The forecasting method combines these reconstruction features with retained blocks, a constrained output decoder, and independent calibration. In the main experiments, histories are standardized using training means and standard deviations. Section 8 separately compares training-based range scaling and bounded raw histories. The input subsample and the maximum number of local training points are fixed by each configuration. Once fitted, the feature maps remain fixed during decoder fitting and calibration.

At stage s, retain the first s blocks:

$$\Gamma_s(x) = (\Phi_1(x)^{\top}, \ldots, \Phi_s(x)^{\top})^{\top} \in \prod_{b=1}^{s} \Delta_{C_b} \subset \mathbb{R}^{p_s}, \qquad p_s = \sum_{b=1}^{s} C_b.$$

A *prefix* means these first s blocks. Their cluster partitions need not be nested: preserving earlier forecasts requires retaining the earlier coordinate functions, not their partitions. The experiments append blocks of sizes $2, 4, 8, 16$, giving prefix dimensions $2, 6, 14, 30$.

3.4 A constrained multihorizon decoder

Let (x_i, Y_i), $i = 1, \ldots, N$, be the training histories and their joint targets. Form the response matrix $\mathsf{Y} \in \mathbb{R}^{N \times q}$ with row i equal to $Y_i^{\top}$, and the feature matrix $F = [F_1, \ldots, F_s] \in \mathbb{R}^{N \times p_s}$, where row i of $F_b \in \mathbb{R}^{N \times C_b}$ is $\Phi_b(x_i)^{\top}$. Fit an intercept $a \in \mathbb{R}^q$ and coefficient matrix $W \in \mathbb{R}^{p_s \times q}$ by solving

$$\min_{a, W} \ \frac{1}{N} \|\mathsf{Y} - \mathbf{1}_N a^{\top} - FW\|_{\mathrm{F}}^2 + \lambda \|W\|_{\mathrm{F}}^2 \quad \text{subject to} \quad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \le \kappa. \tag{2}$$

Here $W_b \in \mathbb{R}^{C_b \times q}$ contains the coefficient rows for block b, $\lambda \geq 0$ is the ridge parameter, and $\kappa \geq 0$ is the coefficient budget. The intercept is unpenalized. The loss sums squared errors over all target coordinates and horizons, then divides by N; it is not additionally divided by q or H.

For a known output box $\mathcal{Y} \subset \mathbb{R}^q$ containing the target, the reported prediction is

$$f_s(x) = \Pi_{\mathcal{Y}}\big(a + W^\top \Gamma_s(x)\big).$$

The objective in (2) fits the unprojected affine predictor; clipping is applied at prediction time and included in every reported loss. Throughout the paper, the same affine map is written as $a + \sum_b W_b^\top \Phi_b(x)$, with $W_b \in \mathbb{R}^{C_b \times q}$.

Centering the training matrices eliminates the intercept. Projected accelerated gradient then solves the constrained quadratic. Each projection computes the block Frobenius norms, projects that nonnegative vector onto an ℓ_1 ball of radius κ, and rescales the blocks. The recorded first-order optimality gap bounds the remaining training-objective error; a small change between iterates alone would not give that bound. Appendix A derives both the projection and the gap. The fitted-risk certificate remains valid after early stopping, provided the returned model satisfies the budget used for certification.

The ridge term regularizes fitting, whereas the block constraint defines the feasible sensitivity class. Exact block duplication leaves that class unchanged, but distributing coefficients across duplicates can reduce the squared Frobenius ridge penalty. The replication results therefore concern feasible prediction classes and their population obstruction. The duplicate-block experiment uses $\lambda = 0$ to compare these feasible classes without changing the fitting penalty.

3.5 Calibration and model selection

The candidate family includes history lengths, dictionary prefixes, geometric feature types, and baseline models. Each certificate compares a representation lower bound with a fitted-model upper bound for the same observed target, origin law, and decoder sensitivity budget. Algorithm 1 summarizes the procedure; Section 4 gives the score formulas and their guarantees.

A lower endpoint above tolerance can motivate changing the representation or budget, but does not establish that a particular change will help. Paired episode losses therefore compare old and new fitted forecasts directly. They can certify an improvement even when the representation obstruction is zero. Conversely, the lower bound can exclude a restricted class even if no tested refinement attains the desired accuracy.

In the experiments, cost is the number of retained feature coordinates, with ties resolved by history length and then validation error. This measures representation size. A deployment with unequal feature-evaluation costs can use a different cost function, provided it is fixed before calibration.

4 Pairwise obstructions and certified forecasting

This section establishes lower bounds on the risk achievable from a fixed representation, upper bounds for fitted forecasts, and conditions for representation refinement. All results refer to the same distribution of histories and observed forecasting targets. In particular, they do not require that the histories form a Markov state, that the trajectory distribution is stationary, or that the learned coordinates satisfy an exact linear evolution equation.

Algorithm 1: Learning and certifying KAHM forecasts

1. Specify horizons, causal origins, known target bounds, RMSE tolerances $\epsilon_h \geq 0$, total failure probability $\delta \in (0, 1)$, and a candidate-cost rule.
2. Use development data to fit geometric blocks and constrained decoders for their retained prefixes. Fit comparison models on the same observed targets.
3. Freeze all candidates, preprocessing choices, sensitivity budgets, history-based strata, costs, concentration bounds, and proposed comparisons. Allocate δ across the risk bounds, stratum-mass intervals, alternative partitions, and improvement comparisons.
4. On $n \geq 4$ independent calibration episodes, compute one mean squared loss per episode and horizon. Compute obstruction scores using disjoint episode pairs at matching origins. For stratified calibration, use one prespecified origin per independent episode (or a justified episode-level stratum), pair within strata, and estimate their masses and confidence intervals.
5. Compute simultaneous lower endpoints L_{jh} and upper endpoints U_{jh} for candidate j at horizon h.
6. Accept a fitted candidate when $U_{jh} \leq \epsilon_h^2$ for every required horizon; exclude its representation class when $L_{jh} > \epsilon_h^2$ for any horizon. Return the least costly accepted candidate, if one exists, and the cost interval between the cheapest nonexcluded class and the cheapest accepted fit.
7. Equal finite cost endpoints certify minimum feasible cost among the declared classes. Exclusion of every class certifies infeasibility within that family. Otherwise report the unresolved accuracy or cost decision explicitly.

Figure 1: Forecasting and certification for a finite family fixed before calibration. The same procedure applies to alternative feature constructions. A candidate fitted after inspecting calibration needs fresh evaluation data or a separately justified adaptive guarantee.

4.1 A directly estimable pairwise obstruction

Let $X \in \mathcal{X}$ be a random available history, let $Y \in \mathbb{R}^q$ be its future target, and let $U = \Gamma(X) \in \mathcal{U}$ be a fixed, measurable representation. Here $\mathcal{X}$ and $\mathcal{U}$ denote the history and representation spaces. A measurable pseudometric $d : \mathcal{U} \times \mathcal{U} \to [0, \infty)$ is symmetric, vanishes on the diagonal, and satisfies the triangle inequality; unlike a metric, it may assign zero distance to distinct points. For a sensitivity budget $\kappa \geq 0$, define

$$\mathcal{G}_\kappa(d) = \{g : \mathcal{U} \to \mathbb{R}^q \text{ measurable} : \|g(u) - g(v)\|_2 \leq \kappa d(u, v) \text{ for all } u, v \in \mathcal{U}\},$$
$$R^*_\kappa(\Gamma, d) = \inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}\|Y - g(\Gamma(X))\|_2^2.$$

Thus κ limits how much a forecast can change when its representation changes, and R^*_κ is the best population squared-error risk permitted by that limit. For any predictor $f : \mathcal{X} \to \mathbb{R}^q$, write $R(f) = \mathbb{E}\|Y - f(X)\|_2^2$. The representation, pseudometric, and budget are fixed before the

calibration data are observed. The class $\mathcal{G}_\kappa(d)$ may contain more decoders than the fitted family; a lower bound on its optimal risk therefore also applies to that family. When the representation and pseudometric are clear, we abbreviate $R^*_\kappa(\Gamma, d)$ to R^*_κ and similarly suppress the arguments of the witness defined below.

To define the pairwise statistic, first draw a conditioning variable A. Then draw (X, Y) and (X', Y') independently from the same conditional law given A, and set $U = \Gamma(X)$ and $U' = \Gamma(X')$. Each copy must have the original marginal law of (X, Y). Taking A to be constant gives an ordinary independent pair. With $z_+ = \max\{z, 0\}$, define

$$W_{\kappa,A}(\Gamma, d) = \frac{1}{2}\mathbb{E}\left[\left(\|Y - Y'\|_2 - \kappa d(U, U')\right)_+^2\right]. \tag{3}$$

The expectation includes both the draw of A and the two conditional draws. We call $W_{\kappa,A}$ a *pair witness*, or *obstruction*: it measures target separation that a decoder with budget κ cannot reproduce from the representation separation. The variable A specifies a sampling construction. Arbitrary nearest-neighbor matching need not have this conditional independence property.

Theorem 1 (Pairwise prediction obstruction) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$. Then*

$$R^*_\kappa(\Gamma, d) \geq W_{\kappa,A}(\Gamma, d). \tag{4}$$

For a representation space $\mathcal{U} \subseteq \mathbb{R}^p$ with $d(u, v) = \|u - v\|_2$, where p is its ambient dimension, assume also $\mathbb{E}\|U\|_2^2 < \infty$. Define

$$\sigma^2_{Y|A} = \mathbb{E}\|Y - \mathbb{E}[Y \mid A]\|_2^2, \qquad \tau^2_{U|A} = \mathbb{E}\|U - \mathbb{E}[U \mid A]\|_2^2,$$

where $\sigma_{Y|A}$ and $\tau_{U|A}$ are the nonnegative square roots. These are scalar, averaged conditional variances, rather than functions of a particular value of A. Then

$$W_{\kappa,A}(\Gamma, \|\cdot\|_2) \geq (\sigma_{Y|A} - \kappa\tau_{U|A})_+^2. \tag{5}$$

Both inequalities use the same conditioning variable and evaluation law.

Proof Fix $g \in \mathcal{G}_\kappa(d)$ with finite risk and define the two residuals $r = Y - g(U)$ and $r' = Y' - g(U')$. Since $Y - Y' = r - r' + g(U) - g(U')$, the triangle inequality gives

$$\|Y - Y'\|_2 \leq \|r - r'\|_2 + \kappa d(U, U'), \qquad (\|Y - Y'\|_2 - \kappa d(U, U'))_+ \leq \|r - r'\|_2.$$

Conditional on A, the residuals are independent and identically distributed. Expanding the squared norm therefore yields

$$\begin{aligned}\mathbb{E}[\|r - r'\|_2^2 \mid A] &= 2\mathbb{E}[\|r\|_2^2 \mid A] - 2\mathbb{E}[r \mid A]^\top \mathbb{E}[r' \mid A] \\ &= 2\left\{\mathbb{E}[\|r\|_2^2 \mid A] - \|\mathbb{E}[r \mid A]\|_2^2\right\}.\end{aligned}$$

Square the preceding pointwise bound, take expectations, and divide by two. This gives $W_{\kappa,A} \leq \mathbb{E}\|r\|_2^2 = R(g \circ \Gamma)$. Taking the infimum over g proves (4); predictors with infinite risk satisfy the bound trivially.

For (5), set $a = \|Y - Y'\|_2$ and $b = \|U - U'\|_2$. Pointwise, $a \le (a - \kappa b)_+ + \kappa b$. For a scalar random variable V, the notation $\|V\|_{L^2} = (\mathbb{E}V^2)^{1/2}$ denotes its root second moment under the pairing law. The triangle inequality in L^2 gives

$$\|(a - \kappa b)_+\|_{L^2} \ge (\|a\|_{L^2} - \kappa\|b\|_{L^2})_+.$$

The same conditional-independence calculation gives $\mathbb{E}a^2 = 2\sigma^2_{Y|A}$ and $\mathbb{E}b^2 = 2\tau^2_{U|A}$. Substitute these identities, square, and divide by two to obtain (5). ■

Proposition 2 (An exact decomposition of witness slack) *For any admissible decoder g with finite risk, put $r = Y - g(U)$, $r' = Y' - g(U')$, and $v = (\|Y - Y'\|_2 - \kappa d(U, U'))_+$. Then*

$$R(g \circ \Gamma) - W_{\kappa,A} = \mathbb{E}\|\mathbb{E}[r \mid A]\|_2^2 + \frac{1}{2}\mathbb{E}\big[\|r - r'\|_2^2 - v^2\big]. \tag{6}$$

Both terms are nonnegative. Consequently, g attains the witness, meaning $R(g \circ \Gamma) = W_{\kappa,A}$, if and only if $\mathbb{E}[r \mid A] = 0$ almost surely and $\|r - r'\|_2 = v$ almost surely. Whenever $r \ne r'$, the latter condition requires $\|g(U) - g(U')\|_2 = \kappa d(U, U')$. It also requires the decoder difference to be a nonnegative scalar multiple of the residual difference; the scalar may be zero.

Proof Conditional independence gives $\frac{1}{2}\mathbb{E}\|r - r'\|_2^2 = R(g \circ \Gamma) - \mathbb{E}\|\mathbb{E}[r \mid A]\|_2^2$. Subtract the witness and use $v \le \|r - r'\|_2$ from the preceding proof. Equality requires both nonnegative terms to vanish. If $r \ne r'$, it also requires equality in $\|Y - Y'\|_2 \le \|r - r'\|_2 + \|g(U) - g(U')\|_2 \le \|r - r'\|_2 + \kappa d(U, U')$, giving the stated conditions. ■

The witness directly estimates a squared distance discrepancy. It does not estimate two variances separately and then subtract their square roots. This distinction can matter when little input variation remains in the representation. The certificates below use concentration for bounded means; they establish no new minimax estimation rate. Taking a square root of a risk bound can still produce slow convergence near zero. Moreover, (5) compares certificates under a common pairing law. It does not compare pairing at a common time origin with conditioning on a different grouping, such as a reference-class label $c(X)$, where c assigns a history to a group.

Proposition 3 (Exactness on a two-atom law) *Suppose (U, Y) takes only two values: (u_0, y_0) with probability $p \in (0, 1)$ and (u_1, y_1) with probability $1 - p$. Use Euclidean representation distance and an unconditioned independent pair, and omit the constant conditioning variable from the notation. Then*

$$R^*_\kappa = W_\kappa = p(1 - p)\big(\|y_1 - y_0\|_2 - \kappa\|u_1 - u_0\|_2\big)^2_+. \tag{7}$$

An affine map attains this optimum. If forecasts must lie in a nonempty closed convex set containing y_0, y_1, projecting that map onto the set preserves both the optimum and the Lipschitz budget.

Proof Write $a = \|y_1 - y_0\|_2$, $b = \|u_1 - u_0\|_2$, and $\mu = py_0 + (1 - p)y_1$. The unequal ordered pairs have total probability $2p(1 - p)$, whereas equal pairs contribute zero. Thus (3) gives the right side of (7) as a lower bound.

To attain the bound, first consider $a = 0$: the constant forecast y_0 has zero risk. If $a > 0$, let $v = (y_1 - y_0)/a$ be the unit target direction and let $s = \min\{a, \kappa b\}$ be the allowed fitted separation. Choose

$$f_0 = \mu - (1-p)sv, \qquad f_1 = \mu + psv.$$

Then $y_0 - f_0 = -(1-p)(a-s)v$ and $y_1 - f_1 = p(a-s)v$, so their weighted squared error is $p(1-p)(a-s)^2$, exactly the proposed bound. If $b = 0$, then $s = 0$ and the constant map μ realizes both fitted values. If $b > 0$, use

$$g(u) = f_0 + \frac{\langle u - u_0, u_1 - u_0 \rangle}{b^2}(f_1 - f_0).$$

Its linear part has operator norm $\|f_1 - f_0\|_2/b = s/b \le \kappa$, so it is an admissible affine decoder. Both fitted values lie on the segment joining y_0 and y_1. Projection onto the stated convex set fixes them and is nonexpansive, meaning that it cannot increase distances. ∎

Exactness also holds on continuous supports. Let T be a square-integrable real random variable, set $U = T$ and $Y = b + \beta T v$, and assume $b, v \in \mathbb{R}^q$, $\|v\|_2 = 1$, and $\beta \ge 0$. Here β is the target's slope, and $\mathrm{Var}(T) = \mathbb{E}(T - \mathbb{E}T)^2$. For independent T, T' and $d(t, t') = |t - t'|$, the pair discrepancy is $(\beta - \kappa)_+|T - T'|$. Since $\mathbb{E}(T - T')^2 = 2\,\mathrm{Var}(T)$, its witness is $(\beta - \kappa)_+^2\,\mathrm{Var}(T)$. The affine decoder

$$g(t) = b + \beta\mathbb{E}[T]v + \min\{\beta, \kappa\}(t - \mathbb{E}[T])v$$

has Lipschitz constant at most κ and residual $(\beta - \kappa)_+(T - \mathbb{E}T)v$. Its risk equals the witness, so $W_\kappa = R^*_\kappa$ also in this continuous-support example.

Two sources of population looseness. Even with independent unconditioned pairs, the witness need not approximate the optimum by a fixed relative factor. Let $\kappa = 1$ and let (U, Y) take values $(3, 0), (0, -1), (1, 1)$ with probabilities $1 - 2\varepsilon, \varepsilon, \varepsilon$, where $0 < \varepsilon < 1/2$. Only the two rare atoms have a positive discrepancy, so $W_\kappa = \varepsilon^2$. The fitted values $g(0) = -1/2$, $g(1) = 1/2$, and $g(3) = 0$ attain risk $\varepsilon/2$. Linear interpolation between these points, with constant values outside $[0, 3]$, defines a 1-Lipschitz decoder. Conversely, any admissible fitted values $v_0 = g(0)$ and $v_1 = g(1)$ satisfy $|v_1 - v_0| \le 1$. The minimum of $(-1 - v_0)^2 + (1 - v_1)^2$ under this constraint is $1/2$, attained at $v_0 = -1/2$ and $v_1 = 1/2$. The two rare atoms alone therefore contribute at least $\varepsilon/2$ to every decoder's risk. Thus $R^*_\kappa = \varepsilon/2$ and $W_\kappa/R^*_\kappa = 2\varepsilon$ can approach zero. The same example lies in Δ_2 after mapping u to $(u/3, 1 - u/3)$ and setting $\kappa = 3/\sqrt{2}$. Stratification below replaces the quadratic probability of encountering this rare pair by the probability of its stratum.

A separate gap comes from comparing the fitted affine family with all Lipschitz decoders. For equally likely $X \in \{-1, 0, 1\}$, take $\Phi(x) = ((x+1)/2, (1-x)/2)^\top$, $Y = X^2$, and $\kappa = \sqrt{2}$, with Euclidean distance between feature vectors. The target satisfies the Lipschitz inequalities on this support and has a scalar Lipschitz extension, so $W_\kappa = R^*_\kappa = 0$. An affine function of $\Phi(x)$ is affine in x; clipping it to $[0, 1]$ preserves monotonicity. A nondecreasing forecast cannot fit the target pair $(1, 0)$ at $x = -1, 0$: its minimum sum of squared errors on that pair is $1/2$, attained when both fitted values are $1/2$. A nonincreasing forecast has the same obstruction on the target pair $(0, 1)$ at $x = 0, 1$. Averaging over the three atoms gives MSE at least $1/6$ for every clipped affine forecast. The unprojected affine optimum is the constant $2/3$, with MSE $2/9$. Thus a sharp lower bound for all Lipschitz decoders need not describe the optimum of a restricted affine family.

4.2 Independent episodes, dependent observations within episodes

An *episode* is one complete observed trajectory, or another complete sampling unit specified by the application. Let $E_1, \dots, E_n$ be independent and identically distributed episodes, also independent of the data used to construct and freeze the models. For horizon $h \in \mathcal{H}$, let $\mathcal{T}_h$ be a fixed, nonempty set of causal forecast origins, available in every episode and used by every compared candidate. Within an episode, observations may be arbitrarily dependent and their distribution may change with time.

Index the J candidates by $j \in \{1, \dots, J\}$ and write $H = |\mathcal{H}|$ for the number of horizons. Candidate j has a representation Γ_j, a representation pseudometric d_j, and a fixed forecaster $f_{jh} = g_{jh} \circ \Gamma_j$ at horizon h. Its decoder g_{jh} has a known Lipschitz budget κ_{jh}. Let $X_{i,t}$ denote the history at origin t in episode i, and let $y_{i,t+h}$ denote the corresponding single-horizon observation (distinct from the joint target Y_t in Section 3). Define

$$Z_{i,jh} = \frac{1}{|\mathcal{T}_h|} \sum_{t \in \mathcal{T}_h} \|y_{i,t+h} - f_{jh}(X_{i,t})\|_2^2, \qquad R_{jh} = \mathbb{E} Z_{i,jh}. \tag{8}$$

Thus $Z_{i,jh}$ is one average loss per episode, and R_{jh} is its population mean. Equivalently, R_{jh} is the risk for a new episode and an independently, uniformly sampled origin from $\mathcal{T}_h$. It is not a guarantee conditional on an individual history. Changing the episode weights or the origin distribution changes the risk being certified.

Assume both targets and predictions lie almost surely in a known nonempty closed convex set $\mathcal{Y}_h$. Let $D_h = \sup_{y,z \in \mathcal{Y}_h} \|y - z\|_2 < \infty$ be its diameter, and put $B_h = D_h^2$. This gives the deterministic loss bound $0 \le Z_{i,jh} \le B_h$. Prediction clipping alone does not establish this bound when the target is unbounded. Nor do training minima and maxima establish a population range.

Form $K = \lfloor n/2 \rfloor$ disjoint episode pairs, leaving one episode unused for pairing when n is odd. For $r \in \{1, \dots, K\}$ define

$$\begin{aligned} Q_{r,jh} = \frac{1}{2|\mathcal{T}_h|} \sum_{t \in \mathcal{T}_h} \Big(& \|y_{2r-1,t+h} - y_{2r,t+h}\|_2 \\ & - \kappa_{jh} d_j\big(\Gamma_j(X_{2r-1,t}), \Gamma_j(X_{2r,t})\big) \Big)_+^2. \end{aligned} \tag{9}$$

For each j, h, the variables $Z_{i,jh}$ are independent across episodes, and $Q_{r,jh}$ are independent across disjoint pairs. Terms within an individual average need not be independent. Their ranges are $[0, B_h]$ and $[0, B_h/2]$, respectively. Write $W_{jh} = \mathbb{E} Q_{r,jh}$. This is exactly the witness in (3) for the horizon-h task when A is a uniformly sampled common origin: conditional on that origin, the two histories and targets come from independent episodes.

For $N \ge 2$ independent, identically distributed scalar observations $V = (V_1, \dots, V_N)$ in a known interval of length $b \ge 0$, define the sample mean and unbiased sample variance by

$$\bar{V} = \frac{1}{N} \sum_{i=1}^{N} V_i, \qquad s_V^2 = \frac{1}{N-1} \sum_{i=1}^{N} (V_i - \bar{V})^2.$$

For a failure probability $\alpha \in (0, 1)$, use the empirical Bernstein radius

$$\mathrm{rad}_{\mathrm{EB}}(V; b, \alpha) = \sqrt{\frac{2 s_V^2 \log(2/\alpha)}{N}} + \frac{7 b \log(2/\alpha)}{3(N-1)}. \tag{10}$$

Here $\log$ is the natural logarithm. The bounded-variable inequality of Maurer and Pontil (2009, Theorem 4), also applied to the reflected observations $-V_i$, gives each of the one-sided bounds $\mathbb{E}V_1 \le \bar{V} + \mathrm{rad}_{\mathrm{EB}}$ and $\mathbb{E}V_1 \ge \bar{V} - \mathrm{rad}_{\mathrm{EB}}$ with failure probability at most α. A joint claim for both directions requires allocation to both events.

Theorem 4 (Simultaneous two-sided episode certificate) *Suppose the J candidates and H horizons are frozen using independent development data, $n \ge 4$, and all fitted decoders satisfy their stated Lipschitz budgets. Fix $\delta \in (0,1)$ and set $\alpha = \delta/(2JH)$. With $\bar{Q}_{jh} = K^{-1}\sum_{r=1}^{K} Q_{r,jh}$ and $\bar{Z}_{jh} = n^{-1}\sum_{i=1}^{n} Z_{i,jh}$, define*

$$L_{jh} = \max\{0, \bar{Q}_{jh} - \mathrm{rad}_{\mathrm{EB}}(Q_{\cdot,jh}; B_h/2, \alpha)\}, \tag{11}$$

$$U_{jh} = \min\{B_h, \bar{Z}_{jh} + \mathrm{rad}_{\mathrm{EB}}(Z_{\cdot,jh}; B_h, \alpha)\}. \tag{12}$$

The dot denotes the full sample over episodes or pairs, so the radius uses sample size K for $Q_{\cdot,jh}$ and n for $Z_{\cdot,jh}$. With probability at least $1-\delta$, conditional on the development data, all candidate–horizon combinations simultaneously satisfy

$$L_{jh} \le W_{jh} \le R^*_{jh} \le R_{jh} \le U_{jh}. \tag{13}$$

Here R^*_{jh} is the infimum risk over all κ_{jh}-Lipschitz decoders of Γ_j, using d_j, the same horizon, the same uniform origin law, and the same raw target.

Proof Condition on the development data so that the entire candidate family is fixed. For each j, h, apply the upper empirical Bernstein bound to the n episode losses and the lower bound to the K pair scores. Since $0 \le W_{jh}$ and $R_{jh} \le B_h$, truncating the endpoints at zero and B_h preserves validity. A union bound over these $2JH$ one-sided events gives $L_{jh} \le W_{jh}$ and $R_{jh} \le U_{jh}$ jointly, with failure probability at most $2JH\alpha = \delta$.

For any admissible decoder, Theorem 1 applies to the joint law obtained by drawing a common uniform origin and two independent episodes. It bounds W_{jh} by that decoder's risk under the same origin mixture. Taking the infimum over these decoders gives $W_{jh} \le R^*_{jh}$. Finally, the fitted decoder belongs to this class, so $R^*_{jh} \le R_{jh}$. The loss and pair statistics may use the same episodes: independence between the upper and lower statistics is unnecessary for the union bound. ■

The probability $1-\delta$ concerns repeated draws of the calibration episodes. The theorem bounds expected squared error; it does not bound every future error by $\sqrt{U_{jh}}$ or provide a prediction interval with conditional coverage. Its independent sample sizes are n and K, not the number of overlapping windows. Additional origins may reduce the variance of an episode average, but they do not create additional independent episodes. The representation, normalization, dictionary sizes, budgets, horizons, and all baselines included in the joint guarantee must be fixed before calibration. They may depend on the development data because the theorem conditions on those data.

Stratification under the same target law. A partition can concentrate pairs on distinctions that unconditional pairing rarely encounters. Condition on the development data throughout this construction, so the representation, metric, and budget are fixed. Fix a measurable map $c : \mathcal{X} \to \{1, \dots, M\}$, with $M \ge 1$, using development data only, and write $\pi_a = \Pr\{c(X) = a\}$. For each

positive-mass stratum, let W_a be the witness from two independent draws conditional on $c(X) = a$. Set $W_a = 0$ when $\pi_a = 0$. The stratified witness is

$$W_{\mathrm{str}} = \sum_{a=1}^{M} \pi_a W_a \le R^*_\kappa. \tag{14}$$

This is Theorem 1 with $A = c(X)$; weighting by the population masses preserves the original target law.

To estimate it, use $n \ge 1$ independent, identically distributed observations (X_i, Y_i) from that law. For each independent trajectory, draw one origin independently from the prespecified common origin law. This produces independent observations under the episode–origin law. Pooling all dependent windows does not. The fitted-risk upper bound may still average all prescribed origins within each episode; a union bound does not require independence between that bound and the stratified statistic. Let $N_a = \sum_i \mathbf{1}\{c(X_i) = a\}$, where $\mathbf{1}\{\cdot\}$ denotes an indicator, and set $K_a = \lfloor N_a/2 \rfloor$. Set $U_i = \Gamma(X_i)$ and write $(U_{a,k}, Y_{a,k})$, $k = 1, \dots, N_a$, for the observations in stratum a, listed in their original sample-index order. For $r = 1, \dots, K_a$, their disjoint pair scores are

$$Q_{a,r} = \frac{1}{2}\left(\|Y_{a,2r-1} - Y_{a,2r}\|_2 - \kappa d(U_{a,2r-1}, U_{a,2r})\right)^2_+ .$$

Assume the target set has squared diameter $B < \infty$, so $0 \le Q_{a,r} \le B/2$.

Theorem 5 (A stratified lower certificate with estimated masses) *Fix $\delta_p, \beta_1, \dots, \beta_M \in (0,1)$ with $\delta_p + \sum_a \beta_a < 1$. If $K_a < 2$, set $\ell_a = 0$; otherwise set*

$$\ell_a = \max\{0, \bar{Q}_a - \mathrm{rad}_{\mathrm{EB}}(Q_{a,\cdot}; B/2, \beta_a)\}.$$

Here $\bar{Q}_a$ is the mean of the K_a scores. Define

$$t_p = \sqrt{\frac{\log(2M/\delta_p)}{2n}}, \qquad p_a^- = (N_a/n - t_p)_+, \qquad p_a^+ = \min\{1, N_a/n + t_p\},$$

and the confidence set, with $\Delta_M = \{p \in [0,1]^M : \sum_{a=1}^M p_a = 1\}$ the probability simplex, $\mathcal{P} = \{p \in \Delta_M : p_a^- \le p_a \le p_a^+ \text{ for every } a\}$. Then

$$L_{\mathrm{str}} := \min_{p \in \mathcal{P}} \sum_a p_a \ell_a \;\le\; W_{\mathrm{str}} \;\le\; R^*_\kappa \tag{15}$$

with probability at least $1 - \delta_p - \sum_a \beta_a$.

Proof Condition on the complete stratum-label vector. The observations in each stratum are independent draws from its conditional law; their disjoint pair scores are therefore independent and identically distributed. For every realized count with $K_a \ge 2$, the conditional failure probability of $\ell_a \le W_a$ is at most β_a. Sparse strata satisfy it deterministically. Averaging over the labels and taking a union bound gives all score inequalities with failure probability at most $\sum_a \beta_a$; there is no additional union over possible counts. For each stratum, Hoeffding's inequality gives $\Pr\{|N_a/n - \pi_a| > t_p\} \le 2\exp(-2nt_p^2) = \delta_p/M$. A union bound over the M strata therefore places $\pi = (\pi_1, \dots, \pi_M)$ in $\mathcal{P}$ with failure probability at most δ_p. On the intersection of these events,

$\min_{p\in\mathcal{P}} \sum_a p_a \ell_a \le \sum_a \pi_a \ell_a \le \sum_a \pi_a W_a$. No independence between masses and score bounds is used. ■

The set $\mathcal{P}$ is nonempty because it contains $(N_a/n)_a$. Its linear minimum is computed by starting at p^- and assigning the remaining mass, up to each upper bound, in increasing order of ℓ_a. Known population masses instead give $L_{\mathrm{str}} = \sum_a \pi_a \ell_a$ with no mass-estimation allocation. For several frozen partitions or candidate–horizon combinations, allocate failure probability over their score events and mass events before taking the largest lower endpoint. A shared partition needs only one mass event. Sparse strata contribute no positive evidence; stratification does not increase the number of independent observations.

Proposition 6 (Exactness on compatible strata) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$. Suppose each positive-mass stratum either has a constant representation $U = u_a$ or a joint law supported on two values of (U, Y). In the first case let its optimal prediction be $\mu_a = \mathbb{E}[Y \mid c(X) = a]$; in the second use the fitted values from Proposition 3, with $b = d(u_0, u_1)$ in that construction. If a single global κ-Lipschitz decoder realizes all these values, then $W_{\mathrm{str}} = R^*_\kappa$. For scalar targets on finite support, global compatibility is equivalent to the pairwise Lipschitz inequalities between all the proposed fitted values, including values from different strata.*

Proof A constant representation gives $W_a = \mathbb{E}[\|Y - \mu_a\|_2^2 \mid c(X) = a]$ by the independent-pair variance identity. A two-atom stratum has the finite-support calculation of Proposition 3. The assumed decoder attains their weighted sum, giving equality in (14). For scalar prescribed values v_i at finitely many support points u_i, define $g(u) = \min_i\{v_i + \kappa d(u, u_i)\}$. Each function in this finite minimum is κ-Lipschitz by the triangle inequality for d, and so is their minimum. At u_j, the term $i = j$ equals v_j and every other term is at least v_j by the assumed pairwise inequalities. Thus $g(u_j) = v_j$. Conversely, a global Lipschitz decoder necessarily satisfies all these pairwise inequalities. ■

For the three-atom example, the stratum $\{3\}$ has zero witness. The stratum $\{0, 1\}$ has mass 2ε, and its two atoms are conditionally equally likely, giving conditional witness $1/4$. Consequently, $W_{\mathrm{str}} = 2\varepsilon(1/4) = \varepsilon/2 = R^*_\kappa$. Stratification can thus remove a population gap, rather than merely improve concentration. It need not improve every witness: singleton strata can remove all pairwise conflicts while incompatible predictions across strata still force positive global risk. Conditional optima cannot be combined without the global compatibility condition.

Corollary 7 (Acceptance, exclusion, and inconclusive decisions) *Let $\epsilon_h \ge 0$ be a prespecified root-mean-square error (RMSE) tolerance at horizon h. On the event in* (13)*, $U_{jh} \le \epsilon_h^2$ certifies that the fitted forecaster attains this tolerance. If $L_{jh} > \epsilon_h^2$, no decoder in the corresponding Lipschitz class attains it. When neither condition holds, the decision is inconclusive. Selecting any member of the frozen family that satisfies $U_{jh} \le \epsilon_h^2$ at every required horizon preserves these guarantees. In particular, one may minimize a predeclared cost over the certified candidates and abstain if there are none.*

Corollary 8 (A certificate of minimum feasible cost) *Let candidate j represent a class $\mathcal{C}_j$ of multihorizon forecasts through Γ_j, containing its fitted forecast and satisfying its stated horizon-specific*

Lipschitz budgets. Give it a fixed cost $c_j \geq 0$. Let c^* be the minimum cost of a class containing a forecast that meets every prescribed tolerance. The minimum of an empty set is $+\infty$. For any simultaneously valid lower bounds L_{jh} on the horizon-h risk of every member of $\mathcal{C}_j$, and upper bounds U_{jh} on its fitted forecast, define

$$\mathcal{E} = \{j : L_{jh} > \epsilon_h^2 \text{ for some } h\}, \qquad \mathcal{A} = \{j : U_{jh} \leq \epsilon_h^2 \text{ for all } h\},$$
$$c_- = \min_{j \notin \mathcal{E}} c_j, \qquad c_+ = \min_{j \in \mathcal{A}} c_j.$$

Thus $\mathcal{E}$ contains the excluded classes and $\mathcal{A}$ contains the accepted fitted candidates. The costs c_- and c_+ refer to the cheapest nonexcluded class and cheapest accepted fit, respectively. On their joint confidence event,

$$c_- \leq c^* \leq c_+. \tag{16}$$

If $c_- = c_+ < \infty$, a least-cost accepted forecast has minimum feasible cost among the specified classes. If every class is excluded, the tolerance requirements are infeasible within their union.

Proof Every accepted fitted forecast meets the tolerances. Every class containing a forecast that meets them is nonexcluded, since one violated horizon lower bound would rule out that forecast. These two inclusions give (16) and both conclusions. A joint coefficient budget can further restrict $\mathcal{C}_j$; exclusion by its larger horizon-wise Lipschitz classes remains valid. ■

If $c_- < c_+$, a cheaper nonexcluded class may still contain an accurate forecast, so selection establishes only the lowest cost among certified candidates. In the experiments, cost counts feature coordinates rather than runtime. Also, (13) implies

$$0 \leq \sqrt{R_{jh}} - \sqrt{R^*_{jh}} \leq \sqrt{U_{jh}} - \sqrt{L_{jh}}.$$

Thus the endpoint gap bounds the fitted decoder's RMSE suboptimality within the stated class. A large gap does not identify optimization error: the witness and either confidence correction may be loose.

For simple sample-size calculations, one may instead prespecify the Hoeffding radius $b\sqrt{\log(1/\alpha)/(2N)}$. If a fitted risk lies at least $\gamma > 0$ below a squared-error tolerance, then

$$n \geq \frac{2B_h^2 \log(1/\alpha)}{\gamma^2}$$

is sufficient for its Hoeffding upper bound to pass that tolerance with probability at least $1 - \alpha$. If the pair witness exceeds the tolerance by γ, then

$$K > \frac{B_h^2 \log(1/\alpha)}{2\gamma^2}$$

is sufficient for the corresponding lower bound to exclude the tolerance with that probability. For acceptance, the stated sample size makes the radius at most $\gamma/2$. A one-sided Hoeffding event bounds the sample mean by the population mean plus $\gamma/2$, so the upper endpoint is at most the tolerance. For exclusion, the strict sample-size inequality makes the radius strictly smaller than $\gamma/2$. On the event $\bar{Q} \geq W - \gamma/2$, where W is the population pair witness and $\bar{Q}$ is its sample mean, the

lower endpoint is therefore strictly above the tolerance. Each event has probability at least $1-\alpha$. These are sufficient sample sizes, not optimal rates. Exclusion depends on the witness margin, rather than on an unknown oracle-risk margin. Choosing the smaller of separately computed Hoeffding and empirical Bernstein bounds requires confidence allocation to both bounds; otherwise the choice must be fixed in advance.

4.3 Refinement that preserves the meaning of the norm budget

Let $\Phi_b : \mathcal{X} \to \Delta_{C_b}$, $b = 1, \dots, s$, be fixed dictionary blocks, where $\Delta_{C_b} = \{u \in \mathbb{R}^{C_b} : u_c \geq 0,\ \sum_c u_c = 1\}$ is the probability simplex. Retain the entire stack $\Gamma_s = (\Phi_1, \dots, \Phi_s)$. Refinement appends a block while keeping every previous coordinate function unchanged. For stacked vectors $u = (u_1, \dots, u_s)$ and $v = (v_1, \dots, v_s)$ define the block maximum metric $\rho_s(u, v) = \max_{1 \leq b \leq s} \|u_b - v_b\|_2$. Its induced pseudometric on histories and the coefficient norm are

$$d_s(x, x') = \rho_s(\Gamma_s(x), \Gamma_s(x')) = \max_{1 \leq b \leq s} \|\Phi_b(x) - \Phi_b(x')\|_2, \qquad \|W\|_{2,1} = \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}}. \tag{17}$$

Here $\|W\|_{2,1}$ is a sum over *blocks*, not over individual rows. As in (2), $a \in \mathbb{R}^q$ and $W_b \in \mathbb{R}^{C_b \times q}$, so the contribution of block b to a column-vector prediction is $W_b^\top \Phi_b(x)$.

Let $\mathcal{Y} \subseteq \mathbb{R}^q$ be a nonempty closed convex output set containing Y almost surely. Write $\Pi_{\mathcal{Y}}(z) = \arg\min_{y \in \mathcal{Y}} \|z - y\|_2$ for Euclidean projection; for a box, it clips each coordinate to its allowed interval. Define

$$\mathcal{F}_{s,\kappa} = \left\{ x \mapsto \Pi_{\mathcal{Y}}\left(a + \sum_{b=1}^{s} W_b^\top \Phi_b(x) \right) : a \in \mathbb{R}^q,\ \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \leq \kappa \right\}. \tag{18}$$

The intercept is unconstrained. Group norm constraints are established tools (Bach, 2008); here the block norm makes the decoder's sensitivity compatible with retained dictionary blocks.

Theorem 9 (Nested and replication-invariant refinement) *Every decoder in $\mathcal{F}_{s,\kappa}$ is κ-Lipschitz with respect to d_s. Appending a block gives $\mathcal{F}_{s,\kappa} \subseteq \mathcal{F}_{s+1,\kappa}$ and $d_{s+1} \geq d_s$ pointwise. Consequently, the infimum prediction risk in this decoder class cannot increase. Neither can the pair witness, provided the budget, target law, and conditional pairing law remain fixed. Appending an exact copy of an existing block leaves both the decoder class and the pair witness unchanged.*

Proof For the unprojected decoder, the triangle inequality and the bound $\|Av\|_2 \leq \|A\|_{\mathrm{F}} \|v\|_2$ give

$$\left\| \sum_b W_b^\top \big(\Phi_b(x) - \Phi_b(x') \big) \right\|_2 \leq \sum_b \|W_b\|_{\mathrm{F}} \|\Phi_b(x) - \Phi_b(x')\|_2 \leq \kappa d_s(x, x').$$

Projection onto a closed convex set is nonexpansive, so this bound also holds for the projected decoder. Assigning zero coefficients to a new block embeds each old decoder in the refined class, proving inclusion and hence nonincrease of the optimal risk. Appending a block can only increase the maximum in d_s. For each paired observation, increasing that distance can only decrease the positive-part discrepancy in (3); taking expectations proves witness monotonicity.

If block $s+1$ exactly copies block b, merge its coefficients into the old block. The prediction is unchanged and $\|W_b + W_{s+1}\|_{\mathrm{F}} \leq \|W_b\|_{\mathrm{F}} + \|W_{s+1}\|_{\mathrm{F}}$, so the merged coefficients remain feasible.

This proves the reverse class inclusion for duplication. A duplicate also leaves the maximum distance unchanged, and therefore leaves the witness unchanged. ■

These claims concern population optima over nested feasible classes, not the risks of separately fitted finite-sample estimators. They also do not assert that every new block contains useful predictive information. Since the Euclidean diameter of each probability simplex is at most $\sqrt{2}$, $d_s \le \sqrt{2}$ regardless of the number of retained blocks. Certifying a fitted decoder at its smaller actual norm is possible, but changes the comparator class. Refinement monotonicity is asserted only at a common budget.

The choice of norm matters even when no information is added. Let two equally likely histories satisfy $\Phi(x_0) = e_1 = (1,0)^\top$ and $\Phi(x_1) = e_2 = (0,1)^\top$, with scalar targets -1 and 1. Under a single Frobenius constraint of radius one, the largest possible forecast separation is $\sqrt{2}$, and the best affine mean squared error (MSE) is $(2-\sqrt{2})^2/4$. After duplicating the block, the concatenated coefficient vector $(-1/2, 1/2, -1/2, 1/2)$ has norm one and predicts both targets exactly. Duplication has relaxed the effective sensitivity constraint. The block-sum constraint prevents this effect while allowing genuinely new blocks to change the geometry.

Theorem 10 (A necessary amount of geometric change) *Let d_0, d_1 be measurable pseudometrics induced on a common history space by an old and a new representation. Use the same square-integrable target, conditionally independent pairing law, and sensitivity budget $\kappa > 0$. Define the random distance increase $\Delta = (d_1(X,X') - d_0(X,X'))_+$ and assume $\|\Delta\|_{L^2} < \infty$. Let W_0, W_1 be the corresponding witnesses from* (3)*, using these induced history distances. Then*

$$\sqrt{W_1} \ge \left(\sqrt{W_0} - \frac{\kappa}{\sqrt{2}} \|\Delta\|_{L^2} \right)_+ . \tag{19}$$

In particular, if a κ-Lipschitz decoder of the new representation attains RMSE at most $\epsilon \ge 0$, then

$$\|\Delta\|_{L^2} \ge \frac{\sqrt{2}}{\kappa} (\sqrt{W_0} - \epsilon)_+ . \tag{20}$$

On any event $0 \le L_0 \le W_0$, replacing W_0 by L_0 on the right side of (20) *is valid simultaneously for every new representation satisfying these conditions. The new representation need not have been fixed when L_0 was computed.*

Proof Set $a = \|Y - Y'\|_2$ and $v_i = (a - \kappa d_i(X,X'))_+$ for $i = 0, 1$. The positive-part function is nondecreasing and 1-Lipschitz. Consequently, $v_0 \le v_1 + \kappa\Delta$ pointwise, and the L^2 triangle inequality gives

$$\|v_1\|_{L^2} \ge (\|v_0\|_{L^2} - \kappa\|\Delta\|_{L^2})_+ .$$

Since $\|v_i\|_{L^2} = \sqrt{2W_i}$, dividing by $\sqrt{2}$ proves (19). If the new decoder has RMSE at most ϵ, Theorem 1 gives $W_1 \le \epsilon^2$. Combining the two inequalities and rearranging gives (20). Finally, its right side is nondecreasing in W_0. Replacing W_0 by a nonnegative lower bound therefore weakens the necessary condition, for every admissible new representation on the same event. ■

This theorem rules out a proposed repair when its population distance increase is too small. It is a necessary condition, not a sufficient refinement rule or a guarantee that an estimated distance

increase is accurate. The quantity L_0 must lower-bound the old *pair witness*; an arbitrary lower bound on old oracle risk cannot be substituted. If the budget also changes from κ_0 to κ_1, the same proof gives the necessary condition

$$\big\|(\kappa_1 d_1(X,X') - \kappa_0 d_0(X,X'))_+\big\|_{L^2} \geq \sqrt{2}(\sqrt{W_{\kappa_0,0}} - \epsilon)_+,$$

where $W_{\kappa_0,0}$ is the old witness at budget κ_0. This form allows the increased separation to come from a larger budget, a changed representation, or both.

4.4 When an added block can improve prediction

The preceding theorem gives a necessary geometric condition. The next calculation gives a sufficient population condition for one proposed feature block: its cross-moment with the current residual must be nonzero, and the old decoder must have unused norm budget. A positive obstruction alone implies neither condition.

Proposition 11 (Residual correlation and budget slack) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$ and consider the unprojected affine predictor*

$$f_0(x) = a + \sum_{b=1}^{s} W_b^\top \Phi_b(x), \qquad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} < \kappa.$$

Let $\psi : \mathcal{X} \to \mathbb{R}^p$ be a new feature block with $\mathbb{E}\|\psi(X)\|_2^2 < \infty$. This block may take general Euclidean values; it need not be simplex-valued. The same sum of block Frobenius norms constrains its coefficients. Define

$$r = Y - f_0(X), \qquad s_0 = \kappa - \sum_b \|W_b\|_{\mathrm{F}} > 0,$$
$$C = \mathbb{E}[\psi(X) r^\top] \in \mathbb{R}^{p\times q}, \qquad G = \mathbb{E}[\psi(X)\psi(X)^\top] \in \mathbb{R}^{p\times p}.$$

Here C is the feature–residual cross-moment, G is the feature second moment matrix, and s_0 is the unused budget. If $C \neq 0$, let $a_0 = \|C\|_{\mathrm{F}}^2$ and $b_0 = \mathrm{tr}(C^\top G C)$, where tr *sums the diagonal entries. Then $b_0 > 0$, and the coefficient block*

$$V = \eta C \in \mathbb{R}^{p\times q}, \qquad \eta = \min\{a_0/b_0,\ s_0/\sqrt{a_0}\}$$

is feasible when appended to the old blocks. Its risk improvement is

$$R(f_0) - R(f_0 + V^\top \psi) = 2\eta a_0 - \eta^2 b_0 \geq \eta a_0 > 0. \tag{21}$$

If Y and $f_0(X)$ belong almost surely to a nonempty closed convex set $\mathcal{Y}$, projecting the new forecast onto $\mathcal{Y}$ retains at least this improvement.

Proof The old simplex-valued blocks are bounded, so $f_0(X)$ and hence the residual r have finite second moments. The same assumption on the new block makes C and G well-defined by the Cauchy–Schwarz inequality. Also, $b_0 = \mathbb{E}\|C^\top \psi(X)\|_2^2$. If $b_0 = 0$, then $C^\top \psi(X) = 0$ almost surely, implying

$$\|C\|_{\mathrm{F}}^2 = \langle C, \mathbb{E}[\psi(X) r^\top]\rangle_{\mathrm{F}} = \mathbb{E}[r^\top C^\top \psi(X)] = 0,$$

contrary to $C \neq 0$. Thus $b_0 > 0$ and $\eta > 0$.

For any coefficient matrix $V \in \mathbb{R}^{p\times q}$, expand $\|r - V^\top \psi(X)\|_2^2$ and take expectations to obtain

$$R(f_0 + V^\top \psi) = R(f_0) - 2\langle V, C\rangle_{\mathrm{F}} + \mathrm{tr}(V^\top G V).$$

Substituting $V = \eta C$ gives the equality in (21). Since $\|V\|_{\mathrm{F}} = \eta\sqrt{a_0} \leq s_0$, appending this block respects the total budget. Since $\eta \leq a_0/b_0$, the improvement is at least $\eta a_0 > 0$. Finally, projection onto a closed convex set containing Y cannot increase the new squared error. The old forecast is already in that set, so its reference risk is unchanged. ■

The condition on the old forecast matters. An improvement between two unprojected affine predictors does not by itself imply an improvement between their separately clipped versions. The proposition also makes no claim when the old decoder exhausts its budget. Its moments are population moments: replacing them with training estimates does not establish the same population improvement. For fitted, clipped models, we therefore compare their episode losses directly.

Fix a horizon h and two forecasts before collecting $n \geq 2$ new independent evaluation episodes. Let Z_i^{old} and Z_i^{new} be their losses on episode i, averaged over the same origins as in (8). Define

$$D_i = Z_i^{\mathrm{old}} - Z_i^{\mathrm{new}} \in [-B_h, B_h], \qquad \bar{D} = \frac{1}{n}\sum_{i=1}^{n} D_i, \qquad R^{\mathrm{old}} = \mathbb{E}Z_i^{\mathrm{old}}, \quad R^{\mathrm{new}} = \mathbb{E}Z_i^{\mathrm{new}}.$$

For $\alpha \in (0,1)$, the lower empirical Bernstein bound gives, with probability at least $1-\alpha$,

$$R^{\mathrm{old}} - R^{\mathrm{new}} \geq \bar{D} - \mathrm{rad}_{\mathrm{EB}}(D; 2B_h, \alpha). \tag{22}$$

The interval length is $2B_h$, because D_i can be negative. A positive right side certifies a risk reduction for these two fitted predictors; a nonpositive value is inconclusive. For multiple fixed comparisons, allocate failure probability over the entire comparison family.

Sequential refinement can also be adaptive. At stage s, a candidate may depend on all previous results. Conditional on this full past, its evaluation episodes must be new independent draws from the specified evaluation law. Choose stage failure probabilities $\alpha_s > 0$ in advance, with $\sum_s \alpha_s \leq \delta$, and accept a change only when its paired lower bound is positive. Conditional validity at each stage and a union bound then make every accepted improvement valid on one event of probability at least $1-\delta$. This argument does not permit repeated tuning on a fixed calibration set.

Finally, a positive obstruction need not be removable by refinement. If Y is independent of X and takes values $-a$ and a with equal probability, where $a > 0$, then the minimum risk over all predictors based on X is a^2. Under unconditioned independent pairing, a constant representation has pair witness a^2; no representation of the same observed history removes this independent future noise. Conversely, a zero witness need not imply a useful predictor. For example, suppose every episode follows the same deterministic trajectory, but its target changes across forecast origins. Pairing two episodes at the same origin gives identical targets and hence a zero witness. A constant representation still cannot predict the variation across origins; its optimal risk is $\mathbb{E}\|Y - \mathbb{E}Y\|_2^2$ under the origin law. Obstruction, fitted risk, and inconclusive decisions therefore answer distinct questions and should be reported separately.

Law	Representation values u_i	Targets y_i	Stratum labels $c(u_i)$
Collapsed binary	$(0, 0)$	$(-1, 1)$	$(0, 0)$
Two atoms	$(-.6/\sqrt{2}, .6/\sqrt{2})$	$(-1, 1)$	$(0, 0)$
Three-point V	$(-1, 0, 1)$	$(1, -1, 1)$	$(0, 0, 1)$
Five-point zigzag	$(-1, -.5, 0, .5, 1)$	$(-1, 1, -1, 1, -1)$	$(0, 0, 1, 1, 1)$
Partial collapse	$(-1, -1, 0, 0, 1, 1)$	$(-1, .2, -.2, .2, -.2, 1)$	$(0, 0, 1, 1, 2, 2)$
Noisy affine	$(-1, -1, 0, 0, 1, 1)$	$(-1, 0, -.5, .5, 0, 1)$	$(0, 0, 1, 1, 2, 2)$
Rare conflict	$(3, 0, 1)$	$(0, -1, 1)$	$(0, 1, 1)$
Affine mismatch	$(-1, 0, 1)$	$(-.5, .5, -.5)$	$(0, 0, 0)$

Table 2: Prespecified joint supports for the lower-bound comparison. Entry i gives one possible pair $(U, Y) = (u_i, y_i)$. Each row assigns equal probabilities to its entries, except rare conflict, whose probabilities are $(.9, .05, .05)$. Repeated representation values retain different possible targets. Stratum labels depend only on U and are fixed before calibration. Every law uses $|u - u'|$, $\kappa = 1$, and predictions in $[-1, 1]$.

5 Lower-bound comparisons beyond two atoms

The two-atom law in Proposition 3 has an exact population witness. To distinguish a loose population bound from finite-sample uncertainty, we evaluate eight finite-support distributions (Table 2). They include deterministic and noisy targets, histories assigned identical features despite different targets, rare conflicting targets, and a case in which affine decoders are less expressive than general Lipschitz decoders. For every method, R^* is the minimum MSE over the same full scalar 1-Lipschitz class with outputs in $[-1, 1]$. These explicit representations isolate the behavior of the certificates; they do not measure the forecasting accuracy of learned KAHM features.

Protocol and comparators. The protocol was recorded before sampling. Each law uses 300 independent calibration samples at each $n \in \{64, 256, 1024, 4096\}$, totaling 9,600 trials. The four primary methods share each sample and each receive a separate one-sided failure probability 0.05; no joint 95% event across methods is asserted. We compare the disjoint-pair empirical Bernstein (EB) bound in Theorem 4, the range-aware centered bound in Appendix B.2, the dual-certified empirical risk minimization (ERM) bound with Kullback–Leibler (KL) correction in (30), and the stratified certificate in Theorem 5. The primary stratified method estimates the stratum probabilities from the calibration data. Both primary pairwise methods use the generic score range $[0, 2]$, obtained from a target diameter of two. Prespecified secondary analyses tighten this range using the known support, use known stratum masses, or replace the empirical-minimum correction by Hoeffding's bound. The result archive includes every secondary analysis.

The empirical and population optimization programs use the full Lipschitz class, with global extension from the finite support. Feasible primal and dual values bracket each optimum; the largest empirical gap is 4.01×10^{-11}. Analytic checks cover all eight population optima. An unprojected affine optimum is never used as a Lipschitz lower bound: in the mismatch law, the former is $2/9$ and the latter is zero.

Population slack and finite-sample cost. Table 3 reports population quantities separately from mean calibrated endpoints. On partial collapse, stratification raises the population witness from 0.0867 to the exact minimum 0.2533. At $n = 4096$, its mean endpoint is 0.1772, compared with

	Population quantities				Mean lower endpoint, $n = 4096$			
Law	R^*	W	W_{str}	C	Pair	Centered	ERM	Strat.
Collapsed binary	1.0000	1.0000	1.0000	1.0000	0.9306	0.9149	0.9345	0.9237
Two atoms	0.3315	0.3315	0.3315	0.3315	0.3034	0.2807	0.2908	0.3001
Three-point V	0.2222	0.2222	0.1667	0.0160	0.1996	0.0000	0.1890	0.1355
Five-point zigzag	0.5400	0.3800	0.5250	0.0744	0.3408	0.0186	0.4891	0.4442
Partial collapse	0.2533	0.0867	0.2533	0.0000	0.0646	0.0000	0.2178	0.1772
Noisy affine	0.2500	0.0972	0.2500	0.0000	0.0777	0.0000	0.2145	0.1801
Rare conflict	0.0250	0.0025	0.0250	0.0000	0.0000	0.0000	0.0148	0.0059
Affine mismatch	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 3: Finite-support comparison on the same full scalar Lipschitz class. R^* is the minimum MSE, W the unconditional pair witness, W_{str} the prespecified stratified witness, and C the centered population bound. The last four columns average 300 independent calibration samples. Each method separately has nominal one-sided confidence 95%; no joint confidence across columns is asserted. Empirical risk minimization (ERM) uses a dual-certified empirical minimum and bounded-loss Kullback–Leibler (KL) inversion. Primary stratification estimates the stratum masses. All quantities are in MSE units.

0.0646 for unconditional pairing and 0.2178 for empirical minimization. Stratification also improves the five-point and noisy-affine witnesses. It worsens the three-point witness, from 0.2222 to 0.1667, demonstrating that a fixed partition need not help.

Empirical minimization gives the largest mean endpoint on most richer laws; the pair bound is larger on the partially separated two-atom and three-point laws at $n = 4096$. Accounting for a known representation range substantially improves centered calibration when the representation is constant. The two-atom comparison in Section 7 quantifies how treating a known zero variance as unknown can weaken a centered endpoint. Consequently, the relative performance of pairwise and centered certificates depends on the estimator and the structural information supplied to it.

Positive tolerances, coverage, and computation. The fixed RMSE tolerances are 0.10, 0.25, and 0.50 for every law. On partial collapse at $n = 4096$, unconditional pairing excludes tolerance 0.25 in 211 of 300 trials; empirical minimization and primary stratification both exclude it in all 300. Rare conflicts remain difficult: their exact risk is 0.025, but unconditional pairing and primary stratification never exclude tolerance 0.10 at this sample size. Empirical minimization excludes it in all 300 trials. Tightening the stratified score range using the known support raises its exclusion count to 284; additionally supplying known masses raises it to 300. The gain therefore depends on both pairing and available information.

All four primary methods have zero observed coverage violations across the prespecified grid. This simulation does not prove coverage; each law–size cell has only 300 repetitions, and the result archive gives binomial Monte Carlo confidence intervals. Reconstruction of all 76,800 primary and secondary endpoints agrees within 1.12×10^{-16}. At $n = 4096$, medians across the eight law-wise median computation times are 31.3, 34.7, 265.4, and 165.0 μs for pair EB, centered, empirical minimum, and stratified EB, respectively. These single-machine measurements use a precomputed finite-support pair-score table and include sample statistics, confidence corrections, and

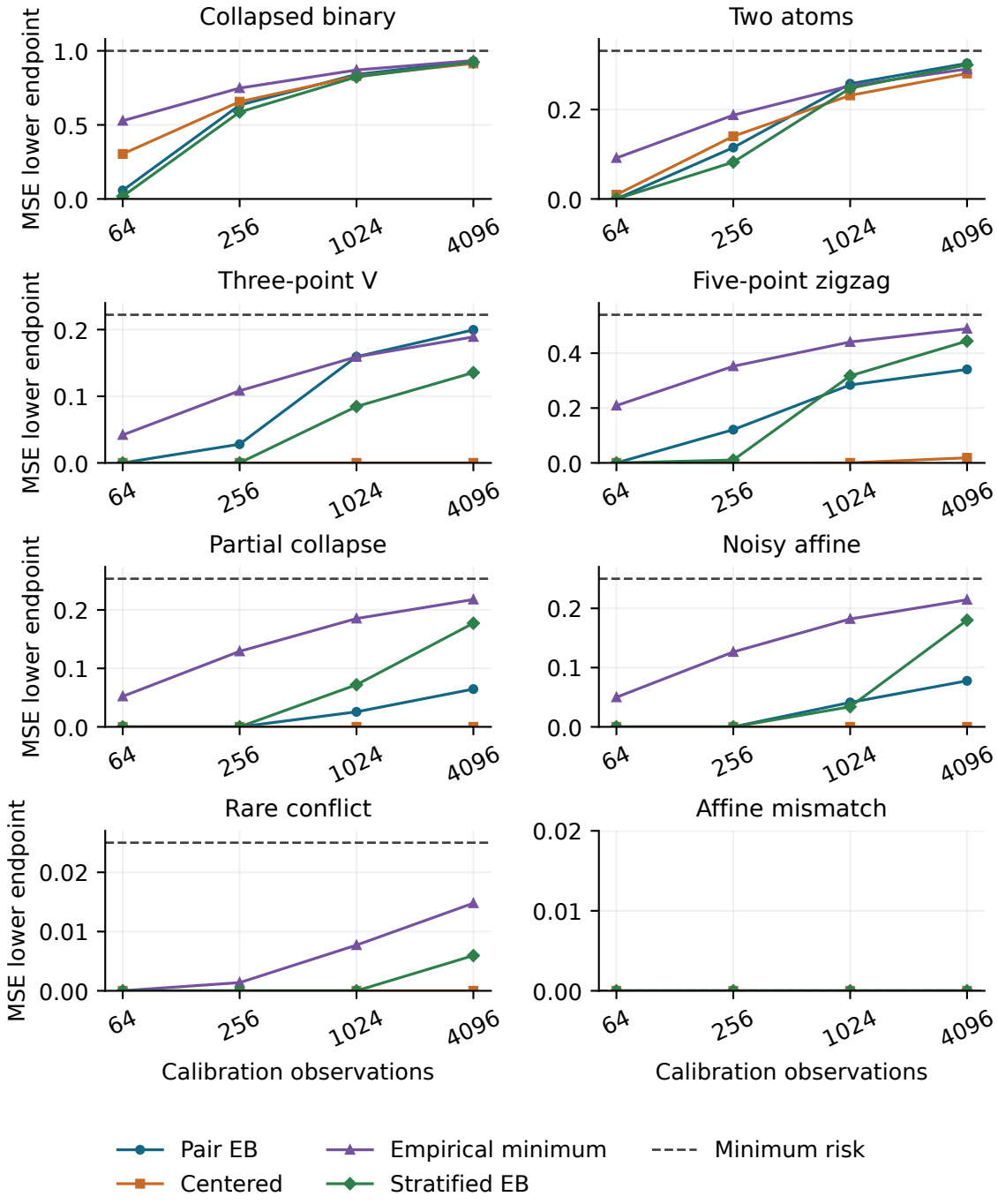


Figure 2: Mean lower endpoints over all 300 repetitions at each calibration size for every prespecified law. Dashed lines show minimum population MSE. Each method has its own 95% one-sided confidence statement. The result archive provides endpoint quantiles, Monte Carlo confidence intervals, secondary methods, and all individual trials.

ERM optimization after representation lookup. They exclude feature learning and do not establish a computational advantage on large continuous supports. The frozen protocol, sufficient statistics, dual certificates, and complete analysis are in `results/witness_comparison`.

6 Decisions at a positive error tolerance

Decisions in the KAHM forecasting studies. The main synthetic, oscillator, and coefficient-budget studies contain 62 fitting runs (Section 7). We distinguish exclusion of a candidate's decoder class from the behavior of a model selected by validation. Among the 800 main-study candidates, the all-horizon decisions at RMSE tolerance 0.35 are 301 accepted, 24 excluded, and 475 unresolved; the corresponding oscillator counts are 55, 30, and 155 among 240 candidates. Nevertheless, all 186 horizon-specific lower endpoints of the 62 models selected by validation across the complete candidate family are zero. Restricting validation selection to retained KAHM gives 184 zero endpoints out of 186. These descriptive counts motivate a separate test of whether lower certificates can resolve a decision that upper calibration leaves open.

A controlled forecasting mechanism. The protocol is specified before calibration observations are generated. Each independent episode supplies one history $X = (Z, S)$ and one scalar future Y. The indicator S equals one with probability 0.2 and zero otherwise, and Z, R, N are mutually independent random signs, each taking -1 and 1 with equal probability, independently of S. We consider two regimes, called repairable and irreducible because the final representation can remove the signal uncertainty only in the first:

$$Y = 0.8SZ + 0.1N \quad \text{(repairable)}, \qquad Y = 0.8SR + 0.1N \quad \text{(irreducible)}.$$

The three entries of (Z, S, Y) occur in temporal order, and the forecast is made after observing Z and S, before observing Y. The target range is $[-0.9, 0.9]$. We append the simplex blocks

$$(1), \qquad (1 - S, S), \qquad \big((1 + SZ)/2, (1 - SZ)/2\big)$$

under the common coefficient budget $\kappa = 2$. The first block has one coordinate; the first two have three coordinates in total; and all three have five. These coordinate counts define the costs $1, 3, 5$ of the three representation classes. These are explicit representations for testing the general theorem; their costs refer only to the three prescribed classes.

At costs 1, 3, and 5, the exact minimum MSEs in the repairable regime are 0.138, 0.138, and 0.01. They all equal 0.138 in the irreducible regime. In the former, the last block represents the conditional mean $0.8SZ$ with coefficients $(0.8, -0.8)$, whose norm is below 2. In the latter, the unobserved sign R leaves the conditional mean zero even with the complete history. The prescribed RMSE tolerance 0.30 therefore requires the last prefix in one regime and is unattainable in the other. For the cost-3 prefix, the unconditional population witness is only 0.0324, whereas pairing conditionally on S gives 0.138. This isolates a population limitation of unconditional pairing, in addition to finite-sample uncertainty.

Calibration and comparisons. Each of 100 independent repetitions per regime uses 2,048 training episodes, 1,024 validation episodes, and nested calibration samples of $256, 1024, 4096, 16384, 65536$ episodes. All three decoders are fitted with zero ridge penalty and frozen before calibration. The RMSE tolerance 0.30 corresponds to an MSE threshold of 0.09. We compare three decision rules:

n	Repairable regime			Irreducible regime	
	Upper acceptance	Minimum cost 5		All classes excluded	
		Unconditional	Stratified	Unconditional	Stratified
256	0	0	0	0	0
1,024	100	0	0	0	0
4,096	100	0	0	0	0
16,384	100	0	100	0	1
65,536	100	0	100	0	100

Table 4: Decision counts out of 100 independent repetitions in each constructed regime, at the prespecified RMSE tolerance 0.30. Upper acceptance certifies a fitted predictor; equality of the cost endpoints certifies the least feasible cost among the three predefined classes. The unconditional and stratified methods share the same upper endpoint and fitted model. Every irreducible-regime upper decision is unresolved. One failure budget 0.05 covers all five calibration sizes and all reported endpoints within each regime and repetition.

accepting when the validation MSE is at most 0.09, accepting when the calibrated upper risk bound is at most 0.09, and combining the same upper bound with unconditional or stratified lower certificates. Validation-only acceptance is a descriptive reference rule without a population confidence guarantee. The two predefined partitions use the history labels S and (S, Z); neither partition uses the future target. For the latter method, we estimate stratum masses and take the maximum of the two separately budgeted endpoints from Theorem 5. One failure budget 0.05 per regime and repetition covers all prefixes and all five sample sizes: 0.02 for upper bounds, 0.01 for unconditional lower bounds, and 0.02 for both stratified families. Calibration uses one forecast per independent episode; it does not treat dependent windows as independent observations.

Decision value and sample requirements. Validation accepts the repaired cost-5 model in all 100 repetitions, and upper calibration does so from $n = 1024$ onward. Stratified lower certificates exclude both cheaper classes in all repetitions at $n = 16384$ and 65536, so the cost interval in Corollary 8 reduces to $[5, 5]$. Unconditional lower calibration never closes this interval at the prescribed sample sizes. Both procedures select the same accepted predictor; the additional conclusion is that no cheaper prescribed class can meet the tolerance, independently of its decoder optimization.

The irreducible control produces no upper acceptance. Stratified lower calibration excludes all three classes in 1 of 100 repetitions at $n = 16384$ and all 100 at $n = 65536$; unconditional lower calibration never excludes all three. Small samples reveal the cost of stratification: at $n = 4096$, unconditional pairing excludes the constant class in every repetition of both regimes, while the stratified method excludes no class. All 600 fits converged, and the exact population risks satisfy every reported simultaneous interval event in all 200 regime–repetition instances. The experiment demonstrates a class-level decision under a controlled mechanism, with substantial sample requirements; it does not establish its frequency in natural forecasting tasks.

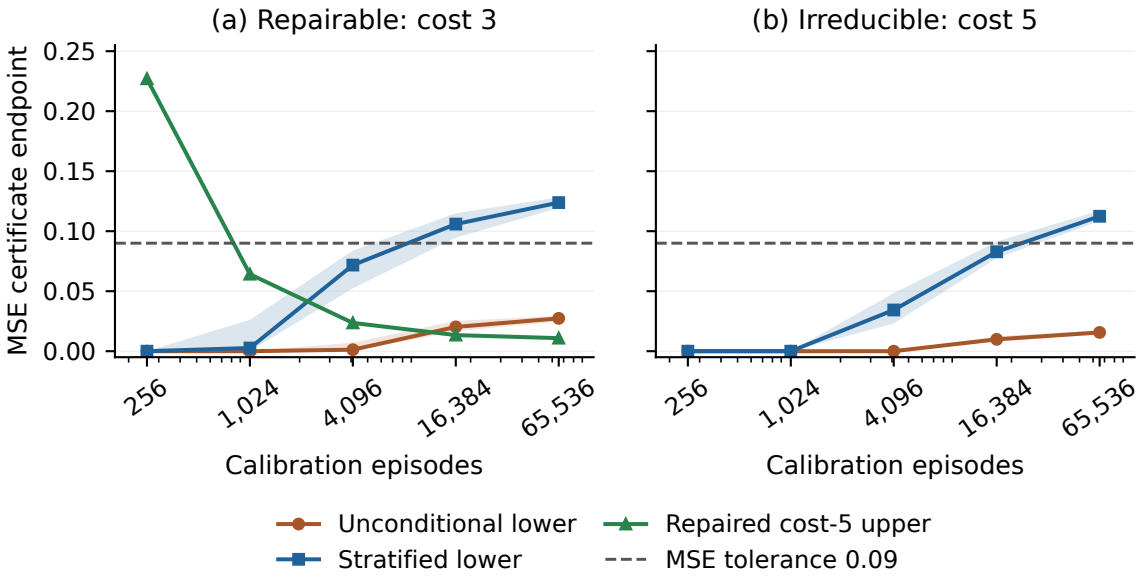


Figure 3: Calibration-size dependence in the constructed decision experiment. Lines show medians and shaded bands the full range across 100 independent repetitions; the bands describe repetition variability. Panel (a) compares lower endpoints for the inadequate cost-3 class with the upper endpoint for the repaired cost-5 predictor. Panel (b) shows lower endpoints for the most detailed class in the irreducible regime. The dashed line is the fixed MSE tolerance 0.09.

7 KAHM forecasting on bounded synthetic systems

The preceding studies test the general risk certificates. This section applies the KAHM learner to four bounded forecasting systems. The comparison includes all prescribed configurations. Empirical test error describes performance on a held-out sample; a certified risk endpoint bounds the corresponding population expectation under the stated sampling assumptions.

7.1 Protocol and predictive accuracy

Four generators produce independent length-72 episodes: a noisy harmonic oscillator, a quasiperiodic signal, nonlinear autoregression with bounded innovations, and a logistic map with an episode-specific parameter. Their observations lie in $[-1, 1]$ by construction (Appendix C). Each of five fitting repetitions uses 128 training, 512 validation, 4096 calibration, and 2048 test episodes. Histories have lengths $m \in \{4, 12\}$, horizons are $\{1, 4, 12\}$, and the 13 common zero-based origins are $11, 15, \ldots, 59$. Within-episode windows may be dependent. Geometry is relearned in each repetition, with budget $\kappa = 4$, ridge coefficient 10^{-4}, affinity exponent 16, and prefix dimensions $2, 6, 14, 30$. Geometry fitting uses at most 1600 histories and 24 local points.

Each repetition freezes 40 candidates: retained KAHM, RBF, and affine-hull features; hard KAHM assignments; ordinary Frobenius-budget and exact-duplicate ablations; ridge autoregression

Method	Harmonic		Quasiperiodic		Nonlinear AR		Logistic	
	$h = 1$	$h = 12$	$h = 1$	$h = 12$	$h = 1$	$h = 12$	$h = 1$	$h = 12$
KAHM blocks	0.070 (0.001)	0.346 (0.006)	0.078 (0.003)	0.284 (0.004)	0.054 (0.001)	0.074 (0.001)	0.176 (0.018)	0.511 (0.001)
RBF blocks	0.068 (0.001)	0.351 (0.003)	0.068 (0.005)	0.287 (0.004)	0.057 (0.001)	0.074 (0.001)	0.172 (0.012)	0.512 (0.001)
Affine-hull blocks	0.072 (0.003)	0.357 (0.004)	0.110 (0.007)	0.275 (0.006)	0.059 (0.003)	0.074 (0.001)	0.156 (0.014)	0.509 (0.001)
Hard KAHM states	0.097 (0.003)	0.358 (0.003)	0.127 (0.002)	0.300 (0.006)	0.062 (0.001)	0.074 (0.001)	0.235 (0.005)	0.512 (0.001)
Ridge AR	0.038 (0.000)	0.281 (0.004)	0.033 (0.000)	0.259 (0.002)	0.047 (0.000)	0.080 (0.001)	0.409 (0.002)	0.507 (0.002)
Quadratic regression	0.031 (0.000)	0.243 (0.003)	0.030 (0.000)	0.246 (0.002)	0.047 (0.000)	0.081 (0.001)	0.010 (0.000)	0.508 (0.003)
KAHM rollout	0.070 (0.002)	0.358 (0.004)	0.079 (0.004)	0.319 (0.011)	0.055 (0.001)	0.074 (0.001)	0.174 (0.018)	0.524 (0.004)
Polynomial rollout	0.030 (0.000)	0.297 (0.010)	0.030 (0.000)	0.288 (0.004)	0.047 (0.000)	0.084 (0.003)	0.010 (0.000)	0.502 (0.004)
Persistence	0.125 (0.000)	1.085 (0.004)	0.084 (0.000)	0.743 (0.004)	0.055 (0.000)	0.103 (0.001)	0.931 (0.006)	0.661 (0.006)

Table 5: Independent-test RMSE: mean over five complete fitting repetitions, with sample standard deviation below in parentheses. One candidate per family is selected by validation MSE across all three horizons before calibration or test scoring. The archive includes $h = 4$, all candidates, and every confidence endpoint. Fitting variation differs from uncertainty conditional on a fitted model; these test errors are descriptive estimates.

(AR); quadratic regression; KAHM and polynomial feature rollouts; persistence; and a constant model. One candidate per family is selected by validation MSE across horizons. These specified configurations assess the procedure, rather than the best possible tuning of each family.

The failure budget 0.05 per repetition allocates 0.04 across $2 \times 40 \times 3$ one-sided risk bounds and 0.01 across 18 adjacent-prefix improvement comparisons. Upper bounds use 4096 episode losses; lower bounds use 2048 disjoint episode pairs. The 13 origins do not increase these independent sample sizes. The joint event is per complete repetition, not across all runs.

Table 5 includes variation across the five fits. KAHM improves substantially over long-horizon persistence on the periodic generators, but quadratic and autoregressive models are stronger there. On nonlinear AR, KAHM horizon-12 RMSE is 0.0743, compared with 0.0802 for ridge AR and 0.0813 for quadratic regression; the latter families are better at horizon one. Direct KAHM fitting improves on feature rollout at horizon 12 on the two periodic generators. The independently initialized oscillators in Section 7.3 show that direct fitting does not always outperform feature rollout.

7.2 Acceptance, exclusion, and fitted refinement

The retained KAHM family certifies the all-horizon RMSE requirement 0.20 for every nonlinear-AR repetition, 0.35 for every quasiperiodic repetition, and 0.50 for every harmonic repetition. Allowing all families also certifies 0.35 on harmonic data. No candidate certifies 0.50 on the logistic task; this is unresolved, not evidence that all true risks exceed the threshold. Even the selected harmonic

	KAHM retained family			All candidates		
System	$\epsilon = .20$	.35	.50	$\epsilon = .20$	.35	.50
Harmonic	0/5 (–)	0/5 (–)	5/5 (6)	0/5 (–)	5/5 (12)	5/5 (4)
Quasiperiodic	0/5 (–)	5/5 (14)	5/5 (2)	0/5 (–)	5/5 (4)	5/5 (1)
Nonlinear AR	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)
Logistic	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)

Table 6: Runs certifying the same RMSE tolerance at all horizons $1, 4, 12$. Parentheses give median retained feature dimension among successful least-dimension selections; '–' means no model passed. Cost is a representation-size proxy, not measured runtime. Each run has a joint 95% event covering its complete frozen family and improvement comparisons.

quadratic model's mean one-step test RMSE 0.0311 corresponds to a mean upper endpoint 0.1497, illustrating the precision cost.

Of 480 retained-KAHM candidate–horizon statements, 119 have positive lower endpoints. Only two exclude RMSE 0.35, and none excludes 0.50; positive endpoints alone are not enough for a useful positive-tolerance decision. Section 6 compares the acceptance, exclusion, and unresolved decisions across the forecasting studies. Paired comparisons certify 72 of 360 adjacent-prefix improvements. Nonlinear AR has no certified adjacent-prefix gain despite meeting useful tolerances. The complete refinement curves and system-specific counts are retained in the repository; nesting a class does not guarantee improvement of each fit.

Exact duplication is already covered analytically by Theorem 9. Zero-ridge numerical checks reproduce matching minimum training risks within their solver gaps. The main duplicate candidate uses zero ridge, whereas primary models use 10^{-4}; their fitted errors are therefore not evidence for ridge-estimator invariance. All 800 main-study candidates completed; the constrained block fits have recorded first-order optimality gaps below 10^{-7}. Independent reconstruction from the stored scores agrees with all 4800 one-sided endpoints and 360 improvement bounds. These checks establish numerical consistency with the specified formulas; statistical validity additionally requires the sampling assumptions in Section 4.

Calibration on a two-atom distribution. A separate benchmark contains 28,800 trials with analytically known class and fitted risks (Appendix D). One pairwise lower endpoint exceeds its true risk; no centered lower or fitted upper endpoint does. The centered estimator in this benchmark estimates representation variance even when that variance is known to be zero. At full collapse and $n = 4096$, its mean lower endpoint is 0.5327. Substituting the known zero variance while keeping the same target correction gives a lower bound of approximately 0.9031 on the expected endpoint; the pairwise method has mean endpoint 0.9177. Thus part of the observed difference is caused by unused structural information. The $n^{-1/4}$ correction of this centered estimator is not a limitation of all centered methods. Section 5 compares a method based on direct standard-deviation concentration and a certified empirical minimum.

7.3 Independently initialized Van der Pol and Duffing episodes

We apply the frozen 40-candidate procedure to independently initialized Van der Pol and forced Duffing episodes, connecting the experiment to Kumar and Kargaran (2026). Each system has three

System	Family	$h = 1$	$h = 4$	$h = 12$	$\overline{\sqrt{U_{12}}}$
Van der Pol	KAHM blocks	0.099 ± 0.001	0.139 ± 0.005	0.381 ± 0.023	0.409
Van der Pol	RBF blocks	0.106 ± 0.005	0.152 ± 0.011	0.383 ± 0.010	0.412
Van der Pol	Affine blocks	0.137 ± 0.003	0.192 ± 0.006	0.299 ± 0.009	0.333
Van der Pol	Ridge AR	0.038 ± 0.000	0.150 ± 0.001	0.353 ± 0.001	0.383
Van der Pol	Quadratic direct	0.032 ± 0.000	0.138 ± 0.001	0.342 ± 0.001	0.374
Van der Pol	KAHM rollout	0.034 ± 0.012	0.064 ± 0.030	0.341 ± 0.033	0.374
Van der Pol	Quadratic rollout	0.029 ± 0.000	0.127 ± 0.002	0.399 ± 0.006	0.427
Van der Pol	Persistence	0.089 ± 0.000	0.339 ± 0.000	0.868 ± 0.001	0.885
Duffing	KAHM blocks	0.080 ± 0.002	0.120 ± 0.004	0.250 ± 0.010	0.300
Duffing	RBF blocks	0.061 ± 0.001	0.098 ± 0.003	0.229 ± 0.007	0.281
Duffing	Affine blocks	0.064 ± 0.004	0.097 ± 0.003	0.223 ± 0.005	0.274
Duffing	Ridge AR	0.019 ± 0.000	0.041 ± 0.000	0.151 ± 0.001	0.214
Duffing	Quadratic direct	0.017 ± 0.000	0.035 ± 0.000	0.137 ± 0.001	0.203
Duffing	KAHM rollout	0.081 ± 0.001	0.126 ± 0.004	0.269 ± 0.014	0.315
Duffing	Quadratic rollout	0.017 ± 0.000	0.035 ± 0.000	0.140 ± 0.002	0.206
Duffing	Persistence	0.033 ± 0.001	0.114 ± 0.002	0.318 ± 0.005	0.363

Table 7: Independent-test sensor RMSE, mean $\pm$ sample standard deviation over three complete fits, with validation selection within each family. The last column averages the selected models' individual horizon-12 upper-RMSE endpoints; this descriptive average gives no ensemble guarantee. All 40 candidates per repetition are archived.

new complete fits with the sample sizes and confidence allocation of Section 7. The bounded position sensor, dynamics, and numerical checks are specified in Appendix C.2. Horizons $1, 4, 12$ correspond to $0.1, 0.4, 1.2$ time units. Certificates concern sensor-output error under the independent-episode numerical law; they require neither stationarity nor independent within-episode windows and do not certify physical-state error.

Retained KAHM improves horizon-12 RMSE over persistence on Van der Pol (0.3814 versus 0.8677) and Duffing (0.2499 versus 0.3185), but other models perform better: affine-hull blocks achieve 0.2990 on Van der Pol, and quadratic regression achieves 0.1369 on Duffing (Table 7). Iterated KAHM achieves 0.3408 on Van der Pol and 0.2686 on Duffing, so direct fitting outperforms iteration only on the latter system.

Retained KAHM certifies all-horizon RMSE 0.35 in all three Duffing fits and 0.50, but not 0.35, in all three Van der Pol fits. Allowing all comparison families certifies 0.35 in every fit of both systems; none certifies 0.20. Adjacent KAHM refinements have positive certified improvement in 43 of 54 Van der Pol comparisons and 10 of 54 Duffing comparisons. The study contains six complete fits and 240 candidates. Reconstructing the 1440 one-sided risk endpoints and 108 paired improvement bounds from the archived scores gives identical numerical values.

7.4 Sensitivity to the coefficient budget

The budget κ defines a feature-dependent class; equal numerical budgets do not imply equal capacity across representations. To assess sensitivity to this choice, a separate experiment uses fresh development, calibration, and test streams with seeds 2701–2703, $m = 12$, and $\kappa \in \{1, 4, 16\}$ on all four

System	κ	Features	$h = 1$	$h = 4$	$h = 12$	Passes
Harmonic	1	16.7	0.3737	0.4102	0.4696	0/3
Harmonic	4	30.0	0.1414	0.2313	0.3900	0/3
Harmonic	16	30.0	0.1392	0.2292	0.3875	0/3
Quasiperiodic	1	24.7	0.2510	0.2905	0.3336	0/3
Quasiperiodic	4	30.0	0.1294	0.2050	0.2927	3/3
Quasiperiodic	16	30.0	0.1294	0.2050	0.2927	3/3
Nonlinear AR	1	7.3	0.1059	0.1056	0.1068	3/3
Nonlinear AR	4	30.0	0.0670	0.0730	0.0746	3/3
Nonlinear AR	16	30.0	0.0670	0.0730	0.0746	3/3
Logistic	1	30.0	0.4713	0.5003	0.5052	0/3
Logistic	4	30.0	0.4384	0.4967	0.5044	0/3
Logistic	16	30.0	0.4384	0.4968	0.5044	0/3

Table 8: Budget sensitivity: mean test RMSE and feature count over three new fitting seeds. "Passes" counts validation-selected KAHM models certified at tolerance 0.35 simultaneously for $h = 1, 4, 12$. The confidence allocation covers all three budgets within each system–seed repetition.

main systems. Other settings match Section 7. The design comprises 36 runs and 756 fitted models, with 21 candidates per budget and system–seed combination. Budgets share streams and geometry within each system–seed repetition. Validation selects a KAHM prefix separately at each budget. Allocating $0.05/3$ to each budget's risk and refinement bounds gives joint confidence at least 0.95 across budgets within a repetition, without a joint claim across all twelve repetitions.

Increasing κ from 1 to 4 reduces one-step KAHM RMSE from 0.3737 to 0.1414 on the harmonic system and from 0.1059 to 0.0670 on nonlinear autoregression; quasiperiodic 0.35 certification increases from zero to three passes. Increasing it from 4 to 16 gives little further improvement: harmonic horizon-12 RMSE changes from 0.3900 to 0.3875, with no 0.35 pass; quasiperiodic and nonlinear-AR errors are unchanged at displayed precision, and logistic horizon-12 RMSE remains about 0.5044, without a pass. This does not establish impossibility at larger budgets or richer geometries.

At $\kappa = 16$, harmonic AR and quadratic baselines achieve horizon-12 RMSEs 0.2377 and 0.2375 and certify 0.35 in all three fits. On nonlinear autoregression at $\kappa = 4$, KAHM has lower horizon-12 RMSE (0.0746 versus 0.0800 and 0.0809) but higher one-step error. Independent reconstruction confirms the 2,268 candidate–horizon certificates and 324 paired improvement bounds. Every constrained fit has a first-order optimality gap below 10^{-7}.

7.5 Writer-disjoint forecasting of original pen trajectories

We use the original, unnormalized PenDigits trajectories (Alpaydin and Alimoglu, 1996); the archive's spatially resampled table uses whole-character preprocessing unsuitable for causal forecasting. The known coordinate range $[0, 500]^2$ permits transforming each coordinate z to $z/250 - 1$ and clipping predictions to $[-1, 1]$, without stroke-specific normalization. Acquisition and development details are in Appendix C.3.

Family	Features	$h = 1$	$h = 2$	$h = 4$
KAHM blocks	30.0	0.0466 ± 0.0010	0.0669 ± 0.0013	0.1127 ± 0.0013
RBF blocks	30.0	0.0421 ± 0.0017	0.0626 ± 0.0019	0.1105 ± 0.0015
Affine-hull blocks	30.0	0.0510 ± 0.0152	0.0664 ± 0.0114	0.1039 ± 0.0053
KAHM, Frobenius budget	30.0	0.0466 ± 0.0010	0.0669 ± 0.0013	0.1127 ± 0.0013
KAHM duplicate blocks	8.0	0.1386 ± 0.0016	0.1532 ± 0.0018	0.1822 ± 0.0018
KAHM hard assignment	16.0	0.0641 ± 0.0004	0.0856 ± 0.0009	0.1297 ± 0.0014
KAHM Koopman rollout	30.0	0.0473 ± 0.0011	0.0688 ± 0.0018	0.1177 ± 0.0025
Ridge autoregression	8.0	0.0130 ± 0.0000	0.0273 ± 0.0000	0.0679 ± 0.0001
Quadratic regression	14.0	0.0077 ± 0.0000	0.0178 ± 0.0001	0.0518 ± 0.0001
Quadratic Koopman rollout	14.0	0.0077 ± 0.0000	0.0185 ± 0.0002	0.0551 ± 0.0005
Persistence	4.0	0.0351 ± 0.0000	0.0696 ± 0.0000	0.1347 ± 0.0000
Constant	1.0	0.2207 ± 0.0007	0.2231 ± 0.0007	0.2273 ± 0.0008

Table 9: Original PenDigits forecasting on 1,751 eligible digits from 13 held-out writers. Coordinate-pooled test RMSE is mean $\pm$ sample standard deviation over three fitting seeds. "Features" is the mean retained count for validation-selected models. Each scalar forecast uses only its own coordinate's history; all selections precede test scoring.

Writer identifiers are inferred from the second integer in each UNIPEN `COMMENT` field: their ranges and record counts agree with the documented 30 development and 14 test writers, but the field semantics are undocumented. We retain the first 32 points of a first pen-down stroke only if its completed length is at least 32. This leaves 1,751 test digits from 13 eligible writers; one of the 14 test writers has no qualifying stroke. The task is retrospective and restricted to writers with eligible strokes; it does not cover every newly started stroke.

Each scalar forecast uses its own coordinate history at horizons $1, 2, 4$. For each of the two coordinates and three fitting seeds, the procedure fits 40 candidates and selects one per family by validation. All candidates and selection rules are frozen before testing. Losses average over origins and digits within writers, then equally over writers and coordinates before taking the square root. Table 9 reports the mean and sample standard deviation over three fitting seeds on the same test cohort; the deviations measure fitting variability, not variation across independent writer cohorts.

Quadratic regression is most accurate at every horizon: its mean RMSEs are 0.00768, 0.01777, and 0.05184, compared with 0.04661, 0.06686, and 0.11269 for retained KAHM. Ridge AR also outperforms KAHM. Compared with persistence, KAHM is worse at one step and better at two and four steps. Retained-block and ordinary Frobenius errors are effectively identical; selected block-norm sums below $\kappa = 4$ are consistent with an inactive constraint.

Population bounds additionally assume eligible writer records are iid from a common population and independent of development; the documentation does not establish this assumption. Writer-averaged scalar losses lie in $[0, 4]$. The prescribed 0.05 allocation over 720 one-sided statements adds 2.42741 to each empirical MSE, clipping upper endpoints at 4 (Appendix C.3). Pooled upper-RMSE endpoints near 1.56 certify none of the tolerances $0.2, 0.35, 0.5$. Neither the 1,751 trajectories nor their overlapping windows increase the independent-writer count. The experiment therefore supports descriptive writer-disjoint comparisons and illustrates limited certification with few groups. Independent reconstruction agrees with all 9,360 candidate–writer–horizon losses to within

1.4×10^{-17}. The fitted coefficient matrices satisfy their budgets, and their first-order optimality gaps are below 10^{-7}. These numerical checks do not establish iid sampling.

8 Input scaling as a model choice

The forecasting experiments in Section 7 standardize histories using training means and standard deviations. To test scale sensitivity, we compare this with training min–max scaling, its clipped version, and raw histories. For a history vector x and coordinate index j, let a_j and b_j be the minimum and maximum of that coordinate over all training histories. The primary alternative transforms the coordinate to

$$z_j(x) = \begin{cases} 2(x_j - a_j)/(b_j - a_j) - 1, & b_j > a_j, \\ 0, & b_j = a_j. \end{cases} \tag{23}$$

Extrema are fixed before geometry subsampling and never recomputed on held-out trajectories. Unseen values may lie outside $[-1, 1]$; the diagnostic clipped version truncates them to this interval. Only KAHM input geometry and models using it change. Targets, budgets, data streams, and non-KAHM controls remain fixed. History and prefix selection use validation MSE within each arm; test errors do not select an arm.

The first comparison reuses every original fitting seed and budget condition. Because this comparison was designed after inspecting the corresponding results, its calibration endpoints do not carry a confirmatory confidence guarantee for the subsequent choice of preprocessing. A fixed confirmation protocol uses new seeds 4701–4703 on all six synthetic systems, with fresh development, calibration, and test episodes. PenDigits instead uses new training subsamples on the same writer cohort; those comparisons remain descriptive.

Table 10 reports all tasks. Mean MSE decreases on harmonic, quasiperiodic, and Duffing data in both stages; Duffing reductions are 7.07% and 7.29%. Logistic MSE increases by 2.65% and 6.97%, and Van der Pol MSE increases by 3.82% and 2.26%. The nonlinear-AR effect changes sign. Mean changes can disagree with a majority of seed-specific changes, so every paired effect and the sample standard deviation across fits are archived alongside conditional episode standard errors. Large numbers of test episodes do not replace independent fitting repetitions.

PenDigits soft-KAHM MSE decreases by 19.79% and 19.65%; all coordinates, horizons, fitting seeds, and held-out writers improve. Ridge AR and quadratic regression remain more accurate. Hard KAHM assignments instead worsen by 3.96% and 4.14% in every fit. Rollout improves on 7 of seven original tasks and 6 of seven new-seed tasks, but the new-seed logistic rollout worsens by 3.61%.

Clipped and unclipped min–max results are similar; raw inputs retain the logistic and Van der Pol limitations. Budget-specific effects also vary. Neither preprocessing nor budget is selected by test performance. All horizon-specific, uncertainty, diagnostic-arm, and budget tables remain in the reproducible result archive. At tolerances $0.20, 0.35, 0.50$, primary scaling leaves the all-horizon upper-certification counts unchanged in every reported synthetic condition. The allocations cover statements within each arm, not selection across arms. Fresh synthetic calibration supports those within-arm claims. Reusing the main-study streams or changing fitting seeds on the same PenDigits cohort does not supply an independent calibration sample for the additional preprocessing choices. The results indicate that the effect of preprocessing depends on both the task and the forecasting architecture.

	Original fitting seeds				New fitting seeds			
Task	Standard	Min–max	$\Delta\%$	Wins	Standard	Min–max	$\Delta\%$	Wins
Harmonic	0.049148	0.048150	-2.03	4/5	0.050240	0.049586	-1.30	3/3
Quasiperiodic	0.037040	0.036182	-2.32	3/5	0.038290	0.035842	-6.39	3/3
Nonlinear AR	0.004561	0.004569	+0.18	4/5	0.004718	0.004656	-1.31	2/3
Logistic	0.164577	0.168931	+2.65	1/5	0.164632	0.176103	+6.97	0/3
Van der Pol	0.058343	0.060572	+3.82	0/3	0.054809	0.056049	+2.26	1/3
Duffing	0.027776	0.025812	-7.07	3/3	0.027913	0.025877	-7.29	3/3
PenDigits	0.006449	0.005173	-19.79	3/3	0.006792	0.005457	-19.65	3/3

Table 10: Validation-selected retained KAHM predictors: mean test MSE over all three horizons and fitting seeds. The primary alternative is training min–max scaling without query clipping. $\Delta\% = 100(\mathrm{MSE}_{\mathrm{minmax}}/\mathrm{MSE}_{\mathrm{standard}} - 1)$; negative values denote improvement. Wins counts fitting repetitions with a smaller all-horizon MSE. Original main tasks have five repetitions; original oscillators and PenDigits have three. All new-seed comparisons have three. Synthetic confirmation uses new development, calibration, and test episodes. PenDigits confirmation uses new training subsamples and the same held-out writer cohort, with coordinate MSE pooled before averaging horizons. Original-seed results reuse previously evaluated streams and are descriptive. No average across different task units is taken.

9 A matched comparison of geometric representations

Design. To separate representation and decoder effects, we compare KAHM, RBF, and affine-hull geometry with three decoders: direct fitting under the block-norm budget in (2), direct fitting under an ordinary Frobenius-norm budget, and feature rollout. We use *head* as a short name for these decoder choices. Every geometry–head pair receives the same prefix dimensions $2, 6, 14, 30$, two history lengths, and four preprocessing options. Training, horizons, origins, regularization, and geometry subsampling follow the original study. Equal numerical budgets and feature counts do not imply equal statistical capacity or computational cost.

Five seeds, 5701–5705, generate fresh episodes for all six synthetic tasks. PenDigits uses new fitting randomness within the same development and held-out writer cohorts. Before calibration, each geometry–head pair selects preprocessing, history, and dimension by validation MSE across horizons, with ties resolved by feature count, history, preprocessing order, and name. PenDigits selects coordinates separately, then averages their losses and horizons within each writer, giving the 13 writers equal weight. Its seed variation describes repeated fitting on that cohort.

Confidence accounting. Within each synthetic task and seed, the four preprocessing arms receive 0.01 each: 0.008 for candidate risk bounds and 0.002 for refinement comparisons. The remaining 0.01 is divided among six primary geometry comparisons. For each, the alternative's loss minus KAHM's loss is averaged within each independent calibration episode over horizons. These differences lie in $[-4, 4]$; the paired empirical Bernstein bound (22) uses range length 8 and tail probability $0.01/6$. The total event has confidence at least 95% within that synthetic task–seed combination. Fixed-dimension and head-selected analyses are descriptive. PenDigits was examined in earlier studies, so

Task	Head	MSE $\times 10^3$: mean (SD) KAHM	RBF	Affine	KAHM change (wins) vs. RBF	vs. Affine
Harmonic	Block	49.77 (1.33)	50.97 (0.98)	52.45 (0.78)	-2.4% (5)	-5.1% (5)
	Frobenius	47.67 (1.31)	48.41 (1.13)	44.94 (1.13)	-1.5% (5)	+6.1% (0)
	Rollout	52.26 (1.04)	52.38 (0.98)	51.66 (1.42)	-0.2% (3)	+1.2% (1)
Quasiperiodic	Block	34.26 (1.32)	34.95 (0.70)	34.74 (2.82)	-2.0% (2)	-1.4% (2)
	Frobenius	33.72 (1.30)	34.30 (0.61)	25.92 (2.72)	-1.7% (3)	+30.1% (0)
	Rollout	39.22 (3.18)	41.75 (3.83)	37.95 (4.58)	-6.1% (2)	+3.3% (1)
Nonlinear AR	Block	4.56 (0.05)	4.69 (0.04)	4.77 (0.04)	-2.7% (5)	-4.4% (5)
	Frobenius	4.56 (0.05)	4.69 (0.04)	4.77 (0.04)	-2.7% (5)	-4.4% (5)
	Rollout	4.57 (0.08)	4.81 (0.07)	4.81 (0.05)	-5.0% (5)	-4.9% (5)
Logistic	Block	165.48 (4.28)	167.18 (2.74)	173.27 (2.22)	-1.0% (3)	-4.5% (5)
	Frobenius	160.66 (5.06)	161.29 (3.38)	170.63 (2.33)	-0.4% (2)	-5.8% (5)
	Rollout	167.63 (5.93)	164.90 (5.64)	177.74 (2.40)	+1.7% (1)	-5.7% (5)
Van der Pol	Block	52.92 (5.56)	61.82 (3.52)	48.35 (3.07)	-14.4% (5)	+9.5% (2)
	Frobenius	26.42 (4.55)	30.92 (7.17)	13.03 (2.92)	-14.5% (3)	+102.7% (0)
	Rollout	33.36 (4.97)	40.23 (16.47)	17.30 (3.63)	-17.1% (4)	+92.8% (0)
Duffing	Block	24.82 (1.60)	23.91 (2.21)	21.95 (2.32)	+3.8% (3)	+13.1% (1)
	Frobenius	23.41 (1.76)	21.21 (2.77)	13.72 (1.51)	+10.4% (0)	+70.6% (0)
	Rollout	27.41 (3.03)	27.02 (4.58)	15.23 (1.41)	+1.4% (3)	+79.9% (0)
PenDigits	Block	5.14 (0.32)	6.09 (0.22)	5.89 (1.15)	-15.6% (5)	-12.7% (4)
	Frobenius	5.04 (0.32)	6.09 (0.22)	5.14 (0.70)	-17.2% (5)	-1.9% (4)
	Rollout	5.46 (0.40)	6.65 (0.34)	5.54 (0.97)	-17.9% (5)	-1.4% (3)

Table 11: Matched geometry comparison after validation-only selection from identical preprocessing, history, and dimension options. Absolute all-horizon test MSE and its sample standard deviation across five fitting repetitions are scaled by 10^3. Relative changes use ratios of mean MSEs; negative values favor KAHM. Wins count repetitions with lower KAHM MSE out of five. PenDigits averages both coordinates and weights the same 13 writers equally. Every effect is descriptive; calibrated comparisons are reported separately.

its follow-up endpoints and paired writer effects provide no new confirmatory population guarantee, even under an iid-writer assumption.

Results and their limits. All 13,440 candidate fits completed. Table 11 combines absolute MSE, its variation across fits, relative changes, and paired seed signs for all 42 contrasts. KAHM has a lower mean MSE in 28 contrasts and a higher mean MSE in 14. These contrasts share models and tasks, so their signs cannot be treated as independent observations. On nonlinear AR, KAHM has lower MSE than both geometric alternatives under every head and seed. Its block-direct MSE 0.00456 also improves on ridge AR 0.00511 and quadratic regression 0.00523.

On PenDigits, KAHM reduces MSE relative to RBF by 15.6%, 17.2%, and 17.9% across the three heads, with lower loss in all five seeds. Its reductions relative to affine geometry are 12.7%, 1.9%, and 1.4%, with four, four, and three favorable seeds. Conventional regressors remain stronger: Frobenius-direct KAHM MSE is 0.00504, compared with 0.00184 for ridge AR and 0.00102 for quadratic regression.

On Van der Pol, KAHM improves on RBF by 14.4–17.1%, but exceeds affine MSE by 9.5%, 102.7%, and 92.8% across the heads. Corresponding Duffing excesses are 13.1%, 70.6%, and 79.9%. Affine geometry has lower MSE than KAHM under Frobenius and rollout heads in every seed on both oscillators. Mean effects can also oppose most paired signs: favorable quasiperiodic block-direct means occur with lower KAHM loss in only two of five seeds.

The complete fixed-dimension curves are released as `paper/figures/matched_geometry_dimensions.pdf`, generated from all fixed-dimension records. On PenDigits, KAHM has lower mean MSE than RBF at every dimension and head; its ranking against affine geometry changes with dimension. On nonlinear AR, KAHM has lower mean MSE than both alternatives at 14 and 30 features, whereas the ranking varies at smaller dimensions. Because histories and preprocessing are reselected, these comparisons are not tests of nesting for fixed blocks.

Only one of the 180 planned synthetic improvement comparisons is certified: Van der Pol seed 5703, KAHM rollout versus RBF rollout, with lower MSE-improvement endpoint 0.00107. All others remain inconclusive. This statement concerns two fitted predictors within the declared event; it is distinct from the descriptive mean effects and implies no population comparison on PenDigits.

10 Speech-energy forecasting with group-level calibration

This second real-data domain tests whether a larger number of documented recording groups permits useful calibration. We use the official Mini Speech Commands subset (Warden, 2018; TensorFlow, 2026), with an eligibility rule, group split, observable, and model grid specified before downloading or inspecting the audio.

Observable and sampling units. Of 8000 source clips, 7178 have the prescribed mono, 16 kHz, signed 16-bit, exactly one-second format. They belong to 1630 filename identifiers, using the source convention before `_nohash_`. All clips from an identifier remain together. A fixed random permutation assigns 244 identifiers to training, 163 to validation, 978 to calibration, and 245 to testing. These contain 1159, 677, 4256, and 1086 clips, respectively. Identifiers denote recording sessions for web volunteers; they are not biometric confirmation of distinct people. Confidence statements assume iid group-record bundles and independence from development data. Identifier separation supports the evaluation design but does not establish that population sampling assumption.

Each clip is divided into 100 nonoverlapping frames of 160 samples. For frame $t \in \{0, \dots, 99\}$ and sample index $j \in \{1, \dots, 160\}$, let x_{tj} be the signed pulse-code modulation (PCM) amplitude divided by 32768. The observed energy is

$$e_t = \frac{10 \log_{10}\left(\max\left\{160^{-1} \sum_{j=1}^{160} x_{tj}^2, 10^{-6}\right\}\right) + 60}{60} \in [0, 1].$$

The implementation clips numerical output to $[0, 1]$. A frame is observed at its end, so this transformation uses no future samples. It represents power in decibels relative to full scale, floored at -60 dB. Multiplying normalized RMSE by 60 gives RMSE in dB for this floored observable, not waveform error or error in unfloored acoustic power. The source selects and quality-screens one-second recordings; our complete-clip eligibility further restricts the population. This is retrospective forecasting within released clips, without per-utterance normalization by future means, peaks, or extrema.

Histories contain 4 or 12 frames, horizons are 1, 4, or 12 frames (10, 40, or 120 ms), and all candidates share origins $11, 15, \dots, 87$. Losses average over origins and clips within each identifier,

Family	Test RMSE in dB: mean (SD) 10 ms	40 ms	120 ms	Maximum upper RMSE in dB 10 ms	40 ms	120 ms	All ≤ 18 dB Fits / 3
KAHM blocks	4.00 (0.15)	7.58 (0.14)	12.73 (0.06)	8.15	11.64	16.37	3/3
RBF blocks	3.79 (0.01)	7.45 (0.01)	12.72 (0.00)	7.83	11.41	16.30	3/3
Affine-hull blocks	4.28 (0.43)	7.77 (0.24)	12.81 (0.05)	8.80	11.98	16.41	3/3
Ridge AR	3.49 (0.00)	7.22 (0.01)	12.51 (0.01)	7.53	11.19	16.15	3/3
Quadratic	3.52 (0.01)	7.24 (0.03)	12.47 (0.03)	7.55	11.21	16.12	3/3
Gaussian kernel	3.51 (0.00)	7.17 (0.01)	12.39 (0.02)	7.57	11.15	16.02	3/3
Persistence	3.56 (0.00)	7.71 (0.00)	14.70 (0.00)	7.58	11.61	18.08	0/3
Constant	15.67 (0.01)	15.63 (0.01)	15.49 (0.01)	19.21	19.17	19.04	0/3

Table 12: Speech-energy forecasting on 245 held-out identifiers. Test means and sample standard deviations use three fits on the same group split. Upper columns take the maximum over those fits, with simultaneous coverage of all 72 selected family–fit–horizon statements under the stated iid group-record assumption. All six learned families pass the 18 dB requirement at every horizon in each fit; none passes the all-horizon 6 or 12 dB requirement. Errors concern the floored energy observable.

then weight identifiers equally. Thousands of clips and windows do not replace the 978 independent calibration units assumed by the theorem.

Development and calibration. Three fitting seeds use the same group split. Each samples 4096 training histories with equal identifier probability and evaluates all validation groups with equal weights. Each fit compares 486 candidates across eight families (Table 12). The geometric families share prefix sizes $2, 6, 14, 30$, standardization or raw inputs, budgets $1, 4$, and ridge parameters $10^{-4}, 10^{-2}$. RBF and affine affinity multipliers are $0.25, 1, 4$; KAHM uses its inherited reconstruction. Linear and quadratic regressions receive four ridge values; Gaussian-kernel regression uses 128 Nyström landmarks, with three bandwidth choices and three ridge values. The repository protocol specifies the complete grids. Every family has both histories. Selection minimizes validation MSE across horizons; all 24 selected models are frozen before held-out losses are evaluated.

For each selected model and horizon, calibration gives one bounded loss in $[0, 1]$ per identifier. The bounded-mean relative-entropy upper bound in Appendix B.3 uses tail probability $0.05/72$, covering all $3 \times 8 \times 3$ statements jointly. Prespecified tolerances of 6, 12, and 18 dB are explicit resolution levels, not claimed application requirements. The reported reproduction evaluates calibration losses and fixes the confidence endpoints before test scoring. The initial execution interleaved calibration and test scoring model by model, so some test losses were available before all calibration endpoints had been frozen. The model choices and calibration formulas were already fixed; rerunning the same procedure in the prescribed order produced identical predictions, group losses, and endpoints. These executions use the same data and constitute one study.

Findings. At 120 ms, mean test RMSE is 12.73 dB for KAHM, 12.51 for ridge AR, 12.47 for quadratic regression, and 12.39 for Gaussian-kernel regression. The latter three also outperform KAHM at shorter horizons. KAHM improves on long-horizon persistence, whose mean is 14.70 dB. These are descriptive accuracy comparisons; no separately calibrated superiority claim is made. The displayed standard deviations describe fitting variation on the same 245 test groups.

All six learned families satisfy the all-horizon 18 dB upper criterion in all three fits. Persistence and the constant model do not; no family passes the all-horizon 6 or 12 dB criterion. Thus the larger calibration cohort can resolve a declared tolerance under the stated sampling model, while leaving tighter requirements unresolved. The experiment supports the certification procedure without establishing that its tolerance is sufficient for an audio application or that KAHM is the best forecaster. Reconstruction from the released records recovers every eligible energy curve, all 88,056 group–model–horizon losses, and all 72 upper endpoints. Shared geometry costs and inference times are archived separately and imply no general computational advantage.

11 Discussion and limitations

What the certificates add. Upper calibration answers whether a fitted forecast meets a tolerance. Lower calibration can exclude every decoder in a specified sensitivity class. Combining the two can certify the minimum feasible representation cost among predefined classes, even when upper-only selection returns the same predictor. The controlled decision experiment demonstrates this distinction, including a case in which all declared classes are excluded. These conclusions concern specified classes and costs; they do not establish globally optimal forecasting complexity or runtime.

Population slack and statistical precision. Proposition 2 identifies two gaps between the witness and a decoder's risk: a nonzero conditional residual mean and a loose geometric inequality for paired residuals. Stratification can repair particular losses but changes the distribution of compared pairs and reduces their counts. It can therefore lower either the population witness or its finite-sample endpoint. The finite-support comparison finds that calibration based on the minimum empirical risk is often tighter and requires more optimization. The pair witness is inexpensive to evaluate, but it need not approximate the optimal risk closely. Its control of the full Lipschitz class can also be less informative than a bound tailored to the fitted affine subclass. Zero obstruction never certifies representation adequacy.

Scope of refinement. Appending feature blocks preserves every earlier feasible forecast. Under the block-norm budget, appending an exact copy of a block leaves the class unchanged. It does not force independently fitted models to improve, and a ridge penalty can change under duplication even when the class does not. When some coefficient budget remains unused, correlation between a new feature block and the prediction residual supplies a sufficient condition for a risk reduction. A separate result gives a necessary amount of representation-distance expansion for attaining a requested tolerance. Neither proves that unsupervised geometric clustering will discover a sufficient state. Experiments freeze candidate families before calibration; sequential refinement on fresh evaluation blocks is a theoretical extension that remains untested.

Independent groups and evaluation populations. Synthetic episodes are independently generated. PenDigits has only 13 eligible held-out writer identifiers, whose interpretation is inferred from source fields; its generic endpoints are uninformative. Speech supplies documented recording identifiers and 978 calibration groups, allowing informative upper bounds under the explicit iid group-record assumption. Source identifiers and a disjoint split do not by themselves establish random population sampling. Both real-data tasks restrict eligibility using completed recordings; the speech source additionally applies clip selection and quality screening. Their risks concern those retrospective acquisition populations. They do not transfer automatically to continuously arriving speech or every

newly started pen stroke. Repeated fitting seeds on one cohort measure fitting variation, not new independent cohorts.

Bounds, adaptation, and distribution shift. The target ranges follow from the synthetic observation laws, the PenDigits acquisition box, or the fixed speech-energy transformation. Clipping predictions alone would not bound squared error for an unbounded target. The certificates do not transfer across changes in dynamics, sensors, noise, or origin laws. The original-seed scaling follow-up reuses calibration after the earlier results motivated its design; its endpoints are nominal. Fresh synthetic confirmation, matched-geometry, and speech studies use the stated independent calibration splits. PenDigits follow-ups reuse the examined writer cohort and remain descriptive. Confidence allocations are stated for each study rather than pooled into one campaign-wide claim.

Accuracy and computational scope. The matched controls support KAHM gains on nonlinear autoregression and against RBF geometry on the fixed PenDigits cohort. Affine-hull alternatives are stronger on the oscillators, while autoregressive, quadratic, and kernel regression outperform KAHM on speech. Scaling yields both gains and regressions. These results support the general certification framework and selected KAHM applications, not universal KAHM superiority. Comparisons use declared implementations and validation grids and do not establish an accuracy frontier against broadly tuned neural forecasters. Feature counts measure representation size. Recorded execution times depend on hardware, shared preprocessing, and implementation; no general speed advantage is inferred.

12 Conclusion

Pairwise prediction obstructions, stratified calibration, and fitted-risk upper bounds can separate achievable accuracy from representation limitations. Retaining feature blocks under a common sensitivity budget makes representation refinements comparable. Joint lower and upper bounds can certify the minimum feasible cost among declared alternatives. Exact examples and comparison bounds identify both the uses and the limitations of the pairwise witness. Forecasting experiments demonstrate informative and unresolved decisions, task-dependent KAHM accuracy, and the importance of defensible independent sampling units. These qualifications are part of the certificate's meaning.

Code, data, and research transparency

The accompanying repository provides the implementation, frozen configurations, result summaries, and figure generators in `extensions/certified_forecasting`; editable manuscript files are in its `paper` directory. The forecasting, input-scaling, matched-geometry, lower-bound comparison, controlled decision, and speech studies have separate protocols and result directories. These provide configurations, source hashes, per-episode scores, and outcomes for every prescribed configuration. Appendix C specifies the forecasting configurations and independent sampling units. The reproduction guide is `extensions/certified_forecasting/docs/REPRODUCIBILITY.md`. Public data are identified by source and checksum; download scripts preserve attribution and license information. Section 10 describes the speech study's evaluation order and its implications for interpretation.

Use of AI tools. AI tools assisted with derivation exploration, implementation, literature retrieval, and manuscript preparation.

Appendix A. Convex fitting and a numerical optimality certificate

This appendix derives the projection used to fit the decoder and a computable upper bound on its training-objective suboptimality. That optimization bound is distinct from the population prediction-risk certificate: one measures how accurately a finite training problem was solved, and the other evaluates forecasting error on new episodes.

Let N be the number of training windows and $p_s = \sum_{b=1}^{s} C_b$ the total number of retained feature coordinates. Write $F = [F_1, \dots, F_s] \in \mathbb{R}^{N \times p_s}$ for the fixed feature matrix, where $F_b \in \mathbb{R}^{N \times C_b}$ contains block b. Let $\mathsf{Y} \in \mathbb{R}^{N \times q}$ contain the corresponding stacked targets, with $q = d_y H$ for d_y output coordinates at each of H horizons. Training windows may be dependent; optimization of this finite objective requires no independence assumption on its rows.

With intercept $a \in \mathbb{R}^q$ and coefficient matrix $W \in \mathbb{R}^{p_s \times q}$, solve

$$\min_{W,a} \frac{1}{N} \|\mathsf{Y} - \mathbf{1}_N a^\top - FW\|_{\mathrm{F}}^2 + \lambda \|W\|_{\mathrm{F}}^2 \quad \text{subject to} \quad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \le \kappa. \tag{24}$$

Here $\mathbf{1}_N \in \mathbb{R}^N$ is the all-ones column vector, $W_b \in \mathbb{R}^{C_b \times q}$ contains the rows associated with F_b, $\lambda \ge 0$ is the ridge coefficient, and $\kappa \ge 0$ is the block-norm budget. The intercept is neither penalized nor constrained. The loss sums over output coordinates and horizons, then averages over training windows; there is no additional division by H or q. Clipping is applied after fitting. Thus (24) minimizes a convex objective before clipping, not the evaluated clipped empirical loss.

Define the row means $\bar{F} = N^{-1}\mathbf{1}_N^\top F$ and $\bar{\mathsf{Y}} = N^{-1}\mathbf{1}_N^\top \mathsf{Y}$, and the centered matrices

$$F_c = F - \mathbf{1}_N \bar{F}, \qquad \mathsf{Y}_c = \mathsf{Y} - \mathbf{1}_N \bar{\mathsf{Y}}.$$

For fixed W, minimizing over the intercept gives $a^\top = \bar{\mathsf{Y}} - \bar{F}W$. Substituting this expression eliminates the intercept. With I_{p_s} the $p_s \times p_s$ identity matrix, put

$$M = F_c^\top F_c / N + \lambda I_{p_s} \in \mathbb{R}^{p_s \times p_s}, \qquad C = F_c^\top \mathsf{Y}_c / N \in \mathbb{R}^{p_s \times q}, \qquad c = \|\mathsf{Y}_c\|_{\mathrm{F}}^2 / N.$$

The reduced objective and its gradient are

$$J(W) = \operatorname{tr}(W^\top M W) - 2\langle W, C \rangle_{\mathrm{F}} + c, \qquad G(W) = \nabla J(W) = 2(MW - C).$$

The Frobenius inner product is $\langle A, B \rangle_{\mathrm{F}} = \operatorname{tr}(A^\top B)$. Since M is positive semidefinite, J is convex. Moreover,

$$\|G(W) - G(V)\|_{\mathrm{F}} \le 2\|M\|_{\mathrm{op}} \|W - V\|_{\mathrm{F}},$$

where $\|M\|_{\mathrm{op}} = \sup_{\|v\|_2 = 1} \|Mv\|_2$ is the operator norm. The implementation uses projected gradient steps with acceleration and monotone restarts. The following formulas assess feasibility and optimization accuracy without requiring a convergence-rate claim for that implementation.

Proposition 12 (Projection onto the block norm ball) *Let $\mathcal{C}_\kappa = \{W : \sum_b \|W_b\|_{\mathrm{F}} \le \kappa\}$ be the feasible coefficient set, and let $A \in \mathbb{R}^{p_s \times q}$ have the same row-block partition as W. Define*

$a_b = \|A_b\|_{\mathrm{F}}$ *and let* $\mathcal{P}_\kappa(A)$ *denote the nearest matrix in* $\mathcal{C}_\kappa$ *in Frobenius norm. If* $\sum_b a_b \le \kappa$*, then* $\mathcal{P}_\kappa(A) = A$*. Otherwise, for* $\kappa > 0$*, choose the unique threshold* $\theta \in (0, \max_b a_b)$ *satisfying* $\sum_b (a_b - \theta)_+ = \kappa$*. For* $\kappa = 0$*, take* $\theta = \max_b a_b$*. The projection is*

$$\big[\mathcal{P}_\kappa(A)\big]_b = \begin{cases} (1 - \theta/a_b)_+ A_b, & a_b > 0, \\ 0, & a_b = 0. \end{cases} \tag{25}$$

In particular, $\mathcal{P}_0(A) = 0$*. For every feasible* W*, the objective gap satisfies*

$$0 \le J(W) - J^* \le \mathcal{D}(W) := \langle G(W), W \rangle_{\mathrm{F}} + \kappa \max_b \|G_b(W)\|_{\mathrm{F}}, \tag{26}$$

where $J^* = \min_{V \in \mathcal{C}_\kappa} J(V)$ *is also the optimum value of* (24)*, and* $G_b(W)$ *is the row block of the gradient corresponding to* W_b*.*

Proof For a block B_b constrained to have Frobenius norm $v_b \ge 0$, Cauchy–Schwarz gives

$$\|B_b - A_b\|_{\mathrm{F}}^2 = v_b^2 + a_b^2 - 2\langle B_b, A_b \rangle_{\mathrm{F}} \ge (v_b - a_b)^2.$$

When $a_b > 0$, equality is attained by taking $B_b = (v_b/a_b)A_b$. When $a_b = 0$, the squared distance is v_b^2 and its optimum has $v_b = 0$. Thus matrix projection reduces to the scalar problem

$$\min_{v_1, \dots, v_s \ge 0} \sum_{b=1}^{s} (v_b - a_b)^2 \quad \text{subject to} \quad \sum_{b=1}^{s} v_b \le \kappa.$$

If the original norms are feasible, the solution is $v_b = a_b$. If $\kappa = 0$, the only feasible vector is zero. In the remaining case the budget constraint is active. A multiplier 2θ for that constraint reduces each coordinate minimization to $\min_{v_b \ge 0} (v_b - a_b)^2 + 2\theta v_b$, whose solution is $v_b = (a_b - \theta)_+$. The sum of these coordinates is continuous and strictly decreasing from $\sum_b a_b$ to zero as θ increases from zero to $\max_b a_b$, so the required threshold exists and is unique. Rescaling each nonzero block gives (25).

The feasible set is nonempty and compact, so continuity of J ensures that J^* is attained. For the objective gap, convexity gives, for every feasible V,

$$J(W) - J(V) \le \langle G(W), W - V \rangle_{\mathrm{F}}.$$

Also,

$$\langle G(W), V \rangle_{\mathrm{F}} \ge -\sum_b \|G_b(W)\|_{\mathrm{F}} \|V_b\|_{\mathrm{F}} \ge -\kappa \max_b \|G_b(W)\|_{\mathrm{F}}.$$

The last value is attainable by assigning the entire budget to a block with largest gradient norm, in the direction opposite to that gradient (or by $V = 0$ if the gradient is zero). Substitute an optimal V in the convexity inequality to obtain (26). ∎

The implementation records coefficient norms, the gap $\mathcal{D}(W)$, iteration count, and whether the requested gap tolerance was reached. These are floating-point diagnostics, not interval-arithmetic guarantees. The held-out risk certificate requires the returned model to satisfy the Lipschitz budget used in calibration; it does not require an optimal training solution.

For a joint multihorizon fit, let $W_{b,h}$ be the columns of block W_b associated with the output at horizon h. Then

$$\sum_b \|W_{b,h}\|_{\mathrm{F}} \le \sum_b \|W_b\|_{\mathrm{F}} \le \kappa,$$

so the joint budget also bounds each horizon decoder's sensitivity. For scalar outputs, $W_{b,h}$ is the column corresponding to h in the increasingly ordered horizon list, and its Frobenius norm is its Euclidean norm. Although the block constraint is invariant to exact feature duplication, the ridge term $\lambda\|W\|_{\mathrm{F}}^2$ is not: splitting coefficients among duplicate blocks changes that penalty. The duplicate-block optimization test therefore uses $\lambda = 0$.

Appendix B. Comparison certificates and bounded-loss calibration

This appendix derives the comparison bounds used in the lower-certificate experiments. Each formula applies to independent, identically distributed calibration units and a model class fixed before calibration. The comparison examines the statistical information and computational cost of these established concentration methods relative to the pair witness.

B.1 Relation to an inverse-contrastive population bound

Let (U, Y) and (U', Y') be independent copies of a representation and its scalar target, with $\mathbb{E}Y^2 < \infty$. Use the same representation pseudometric d and sensitivity budget $\kappa > 0$ in both bounds below. Write W_κ for (3) with constant conditioning variable A. For a threshold $a > 0$, define $F = \{|Y - Y'| > a\}$, $\rho = \Pr(F)$, and $M = \mathbb{E}[\mathbf{1}_F d(U, U')^2] \le \varepsilon < \infty$. Put $v = (|Y - Y'| - \kappa d(U, U'))_+$. Since $\mathbf{1}_F|Y - Y'| \le \mathbf{1}_F v + \kappa \mathbf{1}_F d(U, U')$, the L^2 triangle inequality gives $\|\mathbf{1}_F v\|_{L^2} \ge (\|\mathbf{1}_F|Y - Y'|\|_{L^2} - \kappa\sqrt{M})_+$. On F, the target difference exceeds a, so the first norm is at least $a\sqrt{\rho}$. Therefore

$$\begin{aligned} W_\kappa &\ge \tfrac{1}{2}\|\mathbf{1}_F v\|_{L^2}^2 \ge \tfrac{1}{2}(a\sqrt{\rho} - \kappa\sqrt{M})_+^2 \\ &\ge \tfrac{1}{2}(a\sqrt{\rho} - \kappa\sqrt{\varepsilon})_+^2. \end{aligned} \tag{27}$$

The inverse-contrastive-loss hypothesis of Akash et al. (2021, Lemma 2 and Appendix 7.3) bounds M by ε. In their continuous-target setting and positive regime $\varepsilon < a^2\rho^2/\kappa^2$, their displayed population lower bound is $(a\rho - \kappa\sqrt{\varepsilon})^2/4$; it is no larger than (27) because $\sqrt{\rho} \ge \rho$. Thus the earlier paired-residual argument already implies a prediction-risk obstruction. The positive-part discrepancy gives the stronger population bound above for the same law. A comparison of finite-sample confidence bounds additionally requires the same sampling assumptions and error probability allocations.

B.2 A range-aware centered lower bound

For this scalar comparator, let $U, Y \in \mathbb{R}$ have known interval ranges of lengths D_U, D_Y, and use $d(u, u') = |u - u'|$. From $n \ge 2$ independent, identically distributed observations (U_i, Y_i), define sample means $\bar{U} = n^{-1}\sum_i U_i$ and $\bar{Y} = n^{-1}\sum_i Y_i$, and sample standard deviations

$$s_U = \left(\frac{1}{n-1}\sum_{i=1}^n (U_i - \bar{U})^2\right)^{1/2}, \qquad s_Y = \left(\frac{1}{n-1}\sum_{i=1}^n (Y_i - \bar{Y})^2\right)^{1/2}.$$

Also let $\widehat{v}_Y = n^{-1}\sum_i (Y_i - \bar{Y})^2$, which uses denominator n rather than $n-1$. Define $c_\alpha = \sqrt{2\log(1/\alpha)/(n-1)}$. For a total failure probability $\eta \in (0,1)$, use $\alpha = \eta/3$ if $D_U > 0$ and $\alpha = \eta/2$ if $D_U = 0$, and set

$$\sigma_Y^- = \max\left\{(s_Y - D_Y c_\alpha)_+,\ \sqrt{\left(\widehat{v}_Y - D_Y^2\sqrt{\frac{\log(1/\alpha)}{2n}}\right)_+}\right\},$$

$$\sigma_U^+ = \begin{cases} \min\{s_U + D_U c_\alpha, D_U/2\}, & D_U > 0, \\ 0, & D_U = 0, \end{cases} \qquad L_{\mathrm{ctr}} = (\sigma_Y^- - \kappa\sigma_U^+)_+^2. \tag{28}$$

Then $L_{\mathrm{ctr}} \le R_\kappa^*$ with probability at least $1-\eta$, where R_κ^* is the infimum risk over scalar κ-Lipschitz functions of U. Both target lower bounds hold on the same confidence event, which permits taking their maximum.

To verify this, rescale Maurer and Pontil (2009, Theorem 10) to obtain the one-sided events $\sqrt{\mathrm{Var}(Y)} \ge s_Y - D_Y c_\alpha$ and $\sqrt{\mathrm{Var}(U)} \le s_U + D_U c_\alpha$, each with probability at least $1-\alpha$. Also, $\widehat{v}_Y = \inf_a n^{-1}\sum_i (Y_i - a)^2$, where a may be restricted to the known target interval. Replacing one observation changes this infimum by at most D_Y^2/n, and $\mathbb{E}\widehat{v}_Y = (1-1/n)\,\mathrm{Var}(Y) \le \mathrm{Var}(Y)$. For $D_Y > 0$, the bounded-differences inequality consequently gives

$$\Pr\left\{\widehat{v}_Y - \mathbb{E}\widehat{v}_Y > D_Y^2\sqrt{\frac{\log(1/\alpha)}{2n}}\right\} \le \alpha.$$

This proves the second target lower bound; for $D_Y = 0$ it is identically zero. The deterministic interval bound $\mathrm{Var}(U) \le D_U^2/4$ follows by bounding the mean squared distance to the interval midpoint, and justifies the cap. A union bound combines the two target statements and, when $D_U > 0$, the representation statement. On their common event, (5) with constant A gives $R_\kappa^* \ge (\sqrt{\mathrm{Var}(Y)} - \kappa\sqrt{\mathrm{Var}(U)})_+^2 \ge L_{\mathrm{ctr}}$. When the representation is structurally constant, its variance is exactly zero and requires no confidence correction. A zero sample variance alone does not justify that substitution.

B.3 One-sided relative-entropy bounds

For probabilities $r, p \in [0,1]$, define binary relative entropy

$$\mathrm{kl}(r\|p) = r\log(r/p) + (1-r)\log\{(1-r)/(1-p)\}.$$

Terms with zero numerator are zero. At $p = 0$ or $p = 1$, the value is infinite unless $r = p$, in which case it is zero. For $c > 0$, let $\mathrm{kl}_-(r,c)$ and $\mathrm{kl}_+(r,c)$ be the smallest and largest $p \in [0,1]$ satisfying $\mathrm{kl}(r\|p) \le c$. These endpoints lie respectively below and above r. They are nondecreasing functions of r and can be found by scalar bisection. In particular, $\mathrm{kl}_-(0,c) = 0$ and $\mathrm{kl}_+(1,c) = 1$.

Let $V_1, \ldots, V_n$ be independent and identically distributed in $[0,B]$, with $B > 0$, mean μ, and sample mean $\bar{V}$. For $\eta \in (0,1)$ and $c = \log(1/\eta)/n$, each of

$$B\,\mathrm{kl}_-(\bar{V}/B, c) \le \mu, \qquad \mu \le B\,\mathrm{kl}_+(\bar{V}/B, c) \tag{29}$$

has failure probability at most η. A joint two-sided claim allocates failure probability to both directions. If $B = 0$, both endpoints are zero. The scores need not be Bernoulli observations.

For completeness, convexity of the exponential gives, for every real t,

$$\mathbb{E}e^{tV_1/B} \le 1 - \mu/B + (\mu/B)e^t.$$

Write $p = \mu/B$. Independence and the exponential Markov inequality give, for $z \in (p, 1]$,

$$\Pr\{\bar{V}/B \ge z\} \le \inf_{t \ge 0} e^{-tnz}(1 - p + pe^t)^n = e^{-n\,\mathrm{kl}(z\|p)}.$$

For $z \in [0, p)$, the same calculation with $t \le 0$ bounds $\Pr\{\bar{V}/B \le z\}$ by the same expression (Hoeffding, 1963). If $p \in \{0, 1\}$, boundedness makes every score constant and the assertions hold directly. For $0 < p < 1$, the divergence increases as z moves away from p on either side. Inverting these two one-sided bounds at probability η proves (29). Applying the upper endpoint to losses of a frozen forecaster gives an alternative to (12), under the same sampling and range assumptions. An episode average is one bounded score. Choosing between different concentration methods after seeing calibration requires a joint allocation, just as in Section 4.

B.4 A certified empirical-minimum comparator

An empirical-minimum lower bound provides a comparison with structural certificates that avoid optimization on calibration observations (Kumar and Kargaran, 2026, Section 2.5). Let $\mathcal{G}$ be a fixed nonempty class with losses $\ell_g(X, Y) \in [0, B]$, $B > 0$, and define the following quantities from independent, identically distributed observations (X_i, Y_i) of the target law:

$$R^* = \inf_{g \in \mathcal{G}} \mathbb{E}\ell_g(X, Y), \qquad a_n = \inf_{g \in \mathcal{G}} \frac{1}{n} \sum_i \ell_g(X_i, Y_i).$$

Assume a_n is measurable. Let $\eta \in (0, 1)$ be the allocated failure probability. If a computed measurable value d_n satisfies $d_n \le a_n$ for every calibration sample, put $q_n = \max\{0, d_n/B\}$. Since $0 \le a_n \le B$, one has $0 \le q_n \le a_n/B \le 1$. Then

$$L_{\mathrm{ERM}} = B\,\mathrm{kl}_-\left(q_n, \frac{\log(1/\eta)}{n}\right) \le R^* \tag{30}$$

with probability at least $1 - \eta$.

Indeed, $a_n \le n^{-1} \sum_i \ell_g(X_i, Y_i)$ for every fixed g. For $t \ge 0$, the preceding moment-generating-function bound therefore gives

$$\mathbb{E}e^{tna_n/B} \le \inf_{g \in \mathcal{G}} \{1 - R(g)/B + (R(g)/B)e^t\}^n = \{1 - R^*/B + (R^*/B)e^t\}^n,$$

where $R(g) = \mathbb{E}\ell_g(X, Y)$. The equality follows because $1 - r + re^t$ is nondecreasing in r for $t \ge 0$ and the class has predictors with risks arbitrarily close to R^*. If $R^* = 0$, this moment-generating-function bound and $a_n \ge 0$ imply $a_n = 0$ almost surely. If $R^* = B$, the conclusion follows from $L_{\mathrm{ERM}} \le B$. For $0 < R^* < B$, the exponential Markov argument yields $\Pr\{a_n/B \ge z\} \le \exp\{-n\,\mathrm{kl}(z\|R^*/B)\}$ for $z \in (R^*/B, 1]$. Its inversion bounds $B\,\mathrm{kl}_-(a_n/B, \log(1/\eta)/n)$ from above by R^* with probability at least $1 - \eta$. Monotonicity in the first argument and $q_n \le a_n/B$ prove (30). Neither attainment of the population infimum nor a uniform complexity penalty is needed: the empirical infimum lies below the empirical risk of each fixed population competitor. The result

lower-bounds the risk of every class member; it supplies neither an upper risk guarantee nor an excess-risk bound for a predictor fitted to these calibration observations.

For scalar Lipschitz regression on a finite calibration sample, the empirical infimum is a convex quadratic program (Gottlieb et al., 2017, Section III-A). Restricting predictions to the known target interval $[y_-, y_+]$ preserves the optimum, because clipping reduces loss and preserves the Lipschitz budget. Here $y_- \le y_+$ are known bounds with $Y \in [y_-, y_+]$ almost surely. Let $u_1, \dots, u_{m_U}$ be the $m_U \ge 1$ distinct observed representation values and let $v_j \in [y_-, y_+]$ be the fitted value at u_j. Set

$$w_j = \frac{\#\{i : U_i = u_j\}}{n}, \qquad t_j = \frac{1}{n} \sum_{i:U_i=u_j} Y_i, \qquad \nu = \frac{1}{n} \sum_i Y_i^2.$$

Here $\#$ counts sample indices, $j = 1, \dots, m_U$, and $U_i = \Gamma(X_i)$. Expanding the squared losses shows that the empirical risk is $\nu + \sum_j (w_j v_j^2 - 2 t_j v_j)$. Minimize this expression subject to $v_j - v_k \le \kappa d(u_j, u_k)$ for every ordered pair (j, k) and $y_- \le v_j \le y_+$. These constraints are necessary for a global Lipschitz decoder. They are also sufficient: the function $g(u) = \min_j \{v_j + \kappa d(u, u_j)\}$ is measurable, κ-Lipschitz, and equals v_j at every observed node. Projecting g onto $[y_-, y_+]$ preserves those values and its Lipschitz constant. For ordered scalar inputs with absolute distance, constraints in both directions between adjacent nodes imply every pair constraint by summation.

Let $v = (v_1, \dots, v_{m_U})^\top$ and write the L_{con} pairwise and interval constraints as $Av \le b$, with $A \in \mathbb{R}^{L_{\mathrm{con}} \times m_U}$ and $b \in \mathbb{R}^{L_{\mathrm{con}}}$. For any nonnegative multiplier vector $\zeta \in [0, \infty)^{L_{\mathrm{con}}}$, a dual lower value is

$$d_{\mathrm{dual}}(\zeta) = \nu - \zeta^\top b - \sum_{j=1}^{m_U} \frac{((A^\top \zeta)_j - 2t_j)^2}{4 w_j}. \tag{31}$$

Indeed, adding $\zeta^\top (Av - b)$ to the quadratic objective gives its Lagrangian. Completing the square in each v_j shows that the unconstrained minimum occurs at $v_j = (2t_j - (A^\top \zeta)_j)/(2w_j)$ and equals (31). Every observed node has positive frequency, so $w_j > 0$ and these divisions are defined. For feasible v, the added term is nonpositive. Hence weak duality gives $d_{\mathrm{dual}}(\zeta) \le a_n$, supplying the required d_n. A primal objective alone is an upper bound on a_n and cannot replace this dual value. The implementation records primal feasibility and the primal–dual gap; its floating-point checks do not constitute interval-arithmetic proofs.

Appendix C. Experimental protocols and comparison models

C.1 Episode laws

Each row of a generated data array is one independent episode. In the following definitions, t is an integer time index, y_t is the observed scalar signal, and $\mathrm{Unif}[a, b]$ denotes the uniform distribution on $[a, b]$. All random draws are independent unless an episode-level parameter is explicitly shared across times. The observation range is contained in $[-1, 1]$ for all four main systems. Clipping forecasts to this interval gives scalar squared losses in $[0, 4]$.

1. **Harmonic.** Draw one phase $\theta \sim \mathrm{Unif}[0, 2\pi]$ and one angular frequency $\nu \sim \mathrm{Unif}[.12, .28]$ per episode. Define

$$y_t = .85 \cos(\theta + \nu t) + \eta_t, \qquad \eta_t \sim \mathrm{Unif}[-.03, .03],$$

   where η_t is observation noise. Thus $|y_t| \le .88$.

2. **Quasiperiodic.** Draw phases θ_1 and θ_2 independently from $\mathrm{Unif}[0, 2\pi]$, and draw $\nu \sim \mathrm{Unif}[.10, .22]$ once per episode. Set

$$y_t = .55\cos(\theta_1 + \nu t) + .30\cos(\theta_2 + \sqrt{2}\nu t) + \eta_t, \qquad \eta_t \sim \mathrm{Unif}[-.03, .03].$$

   The two angular frequencies are ν and $\sqrt{2}\nu$. Again $|y_t| \leq .88$.

3. **Nonlinear autoregression.** Initialize y_0 and y_1 independently from $\mathrm{Unif}[-.8, .8]$. For $t \geq 1$, apply

$$y_{t+1} = \tanh\{1.25y_t - .65y_{t-1} + .25\sin(3y_t) + \eta_{t+1}\}, \qquad \eta_{t+1} \sim \mathrm{Unif}[-.10, .10],$$

   where η_{t+1} is an innovation and $\tanh$ denotes the hyperbolic tangent. The implementation discards the first 80 stored values before retaining a length-72 episode. The two initial values occupy indices zero and one; recursive updates start at index two. The hyperbolic tangent ensures the stated output bound.

4. **Logistic.** Draw the episode parameter $r \sim \mathrm{Unif}[3.65, 3.95]$ and initial internal state $u_0 \sim \mathrm{Unif}[.05, .95]$, then iterate $u_{t+1} = ru_t(1 - u_t)$. After 40 burn-in updates, store the next 72 updated values, transformed to the observed scale by $y = 2u - 1$. Because $r \leq 4$, updates preserve $u \in [0, 1]$ and hence $y \in [-1, 1]$. Within-episode dynamics can be chaotic. Independence between episodes follows from independent initialization, without requiring a mixing argument.

The finite generation and origin protocol defines the risk being estimated. No assumption of stationarity after burn-in or sampling from an invariant distribution is needed. Each generator has its own observation scale and horizon interpretation; we do not average errors across generators as if they were a common physical measure.

C.2 Oscillator protocol

For continuous time t, position $x(t)$ and velocity $v(t)$ satisfy

$$\begin{aligned} &\dot{x} = v, && \dot{v} = (1 - x^2)v - x && \text{(Van der Pol)},\\ &\dot{x} = v, && \dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \varphi) && \text{(Duffing)}, \end{aligned}$$

where dots denote time derivatives. Initial position and velocity are independent $\mathrm{Unif}[-2, 2]$ draws; Duffing phase $\varphi \sim \mathrm{Unif}[0, 2\pi]$ is also independent per episode. Fourth-order Runge–Kutta uses step 0.025 and discards ten time units. The 72 sensor observations are

$$y_j = 0.95\tanh(x(10 + 0.1j)) + \eta_j, \qquad \eta_j \overset{\text{iid}}{\sim} \mathrm{Unif}[-0.02, 0.02], \quad j = 0, \ldots, 71.$$

Their range $[-0.97, 0.97]$ lies within the certificate range $[-1, 1]$. A prespecified half-step check on 16 fixed initializations, with phases and noise shared between steps 0.025 and 0.0125, gives maximum sensor changes 1.69×10^{-6} (Van der Pol) and 3.49×10^{-6} (Duffing), below 10^{-3}. This checks numerical consistency without a rigorous continuous-time integration-error bound.

C.3 PenDigits acquisition and development

The original UNIPEN files contain 7,494 development and 3,498 test digits, with nominal 100 ms sampling. Every coordinate is checked against the acquisition range $[0, 500]$; no stroke centering, alignment, or resampling is applied, and pen-up gaps are excluded. Writer identifiers include the source split; approximately 250 records per identifier support the inference described in Section 7.5.

The first 24 development writers supply training and the remaining six validation. Each of three seeds selects 20 eligible strokes per training writer (480 total); validation uses all 717 eligible strokes. Histories have lengths $m \in \{4, 8\}$, common zero-based origins $7, 11, 15, 19, 23, 27$, and horizons $1, 2, 4$. The 40-candidate family follows the main comparison design, with successive block sizes $2, 4, 8, 16$, $\kappa = 4$, ridge $\lambda = 10^{-4}$, and affinity exponent $\omega = 16$. Validation weights writers equally and averages MSE over horizons. Normalized RMSE 0.01 equals 2.5 acquisition pixels. For the additional iid-writer calculation, two coordinates, three fits, 40 candidates, and three horizons give 720 one-sided statements. With 13 writers, the Hoeffding addition is $4\sqrt{\log(720/0.05)/(2 \cdot 13)} = 2.42741$.

C.4 Random streams and configuration

Main fitting seeds are $1701, \ldots, 1705$. For seed s, separate NumPy streams use $100s+11$, $100s+23$, $100s + 47$, and $100s + 89$ for training, validation, calibration, and testing. Clustering and retained blocks receive deterministic offsets from the training seed and history length. Archived configurations specify these offsets and random-generator versions. Geometry uses at most 1600 training rows; decoders use all 1664 rows (128 episodes, 13 common origins). Overlapping training rows are permitted. Validation and test losses weight episodes equally and origins equally within episodes; selection averages MSE over the three horizons.

C.5 Comparison definitions

In the main synthetic campaigns, RBF and affine-hull comparators use the same type of k-means partition, with five initializations and at most 150 iterations. Here k is the number of coordinates in the block. To define their scores, suppress the block index. Let $\xi \in \mathbb{R}^{d_x}$ be a preprocessed history, where d_x is its input dimension. For cluster $c \in \{1, \ldots, k\}$, let $\mu_c \in \mathbb{R}^{d_x}$ be its fitted center, $\mathcal{I}_c$ the nonempty set of assigned geometry-training indices, and $n_c = |\mathcal{I}_c|$. Define the cluster's squared-radius scale and the global RBF scale by

$$s_c = \max\left\{\frac{1}{n_c}\sum_{i\in\mathcal{I}_c}\|\xi_i - \mu_c\|_2^2,\ 10^{-6}\right\}, \qquad \beta = \mathrm{median}_{1\le c\le k}\, s_c,$$

where median is the usual sample median. The RBF score is

$$a_c^{\mathrm{RBF}}(\xi) = \frac{\|\xi - \mu_c\|_2^2}{2\beta}.$$

Thus every cluster uses the same bandwidth, while the positive floor prevents division by a zero radius.

For the affine-hull comparator, take the singular value decomposition of the matrix whose rows are $(\xi_i - \mu_c)^\top$, $i \in \mathcal{I}_c$. When $n_c \ge 2$, let V_c contain its first $k_c = \min\{2, d_x - 1, n_c\}$ right singular vectors as columns. For a singleton cluster, set $k_c = 0$ and use an empty basis. Then $P_c = V_c V_c^\top$ is

the orthogonal projection onto the selected local subspace; $P_c = 0$ for an empty basis. The constraint $k_c \le d_x - 1$ leaves an orthogonal direction whenever one is available. Define

$$r_c = \max\left\{\frac{1}{n_c}\sum_{i\in\mathcal{I}_c}\|(I_{d_x} - P_c)(\xi_i - \mu_c)\|_2^2,\ 0.05 s_c,\ 10^{-6}\right\},$$
$$a_c^{\mathrm{AH}}(\xi) = \frac{\|(I_{d_x} - P_c)(\xi - \mu_c)\|_2^2}{r_c} + 0.1\frac{\|\xi - \mu_c\|_2^2}{s_c},$$

where I_{d_x} is the identity matrix. The first term measures distance from the local affine subspace; the second penalizes distance from its center. The floors in r_c keep the normalization positive even when the training residuals vanish.

For either choice $a_c = a_c^{\mathrm{RBF}}$ or $a_c = a_c^{\mathrm{AH}}$, the block coordinates are

$$\Phi_c(\xi) = \frac{\exp(-a_c(\xi))}{\sum_{j=1}^{k}\exp(-a_j(\xi))}.$$

Normalization is evaluated in log space. The score multiplier is one in the original synthetic campaigns; subsequent studies specify their expanded grids. Both alternatives use the retained-block decoder.

Hard KAHM states replace the final block by a one-hot vector: its largest coordinate becomes one and all others become zero. Ridge AR uses the original causal history. Quadratic regression uses training-standardized histories and all monomials of degree one or two, including cross-products; it does not add a constant feature because the intercept is unpenalized. Both direct decoders use the stated Frobenius constraint and ridge coefficient. Persistence repeats the latest observation at every horizon. The constant model predicts the training target mean separately for each horizon and has sensitivity budget zero.

For the iterated baselines, let $z_t \in \mathbb{R}^{1\times p}$ be the row vector of p features at time t. Ridge regression fits an affine feature transition and a scalar observation decoder,

$$z_{t+1} \approx z_t M + b, \qquad y_t \approx z_t v + c,$$

where $M \in \mathbb{R}^{p\times p}$ is the transition matrix, $b \in \mathbb{R}^{1\times p}$ is its intercept, $v \in \mathbb{R}^p$ is the observation coefficient vector, and $c \in \mathbb{R}$ is the observation intercept. For a positive integer horizon h, composing these fitted maps gives the un-clipped forecast

$$\widetilde{y}_{t+h} = z_t M^h v + b\left(\sum_{\ell=0}^{h-1} M^\ell\right) v + c,$$

with M^0 the identity matrix. Only this final scalar forecast is clipped. The effective output coefficient is $M^h v$; its Euclidean norm is the frozen Lipschitz budget relative to Euclidean feature distance at horizon h. The certificate evaluates this fitted forecast directly and requires no spectral-stability assumption, exact feature closure, or approximation to the trajectory distribution.

C.6 Reproduction and study provenance

The repository provides configurations, source hashes, candidate definitions, per-episode scores, and scripts for independently reconstructing the reported confidence endpoints. Episode-score

archives support reconstruction; fixed-seed runners regenerate large files omitted from the repository. Archived environments specify numerical-library versions; bitwise agreement across architectures is not assumed. Complete commands for all studies, asset generation, and manuscript compilation are given in `extensions/certified_forecasting/docs/REPRODUCIBILITY.md`. Runners preserve failed attempts and refuse to overwrite completed runs. Original-seed and confirmation input-scaling analyses remain separate, with the scope in Section 8.

C.7 Matched-geometry protocol

Each preprocessing arm has 72 matched scalar candidates and 12 controls. The refinement allocation covers 54 adjacent-prefix horizon comparisons: three geometries, two histories, three adjacent prefix pairs, and three horizons. The complete schedule and configuration hashes precede fitting; Section 9 specifies selection and joint confidence allocation. The result archive contains all 13,440 candidates, paired seed effects, horizon-specific results, and PenDigits writer comparisons. Table 11 combines absolute errors, fitting variation, and relative effects. The asset-generation scripts read these complete analysis records without refitting models or selecting configurations. Each study has a separate configuration and provenance record.

Appendix D. Exact-risk benchmark and centered comparator

This benchmark has an analytically known optimal risk and fitted-model risk, so certificate coverage can be checked without using a test estimate as population truth. Let $Z \in \{0, 1\}$ be a Bernoulli state with $\Pr(Z = 1) = p \in (0, 1)$, and let the scalar target be $Y = 2Z-1$. For a representation-separation parameter $a \in [0, 1]$, use

$$\Phi_a(0) = ((1 + a)/2, (1 - a)/2)^\top, \qquad \Phi_a(1) = ((1 - a)/2, (1 + a)/2)^\top.$$

Both vectors lie in the two-coordinate probability simplex. Their Euclidean distance is $\sqrt{2}a$, whereas the target separation is 2. For sensitivity budget $\kappa \geq 0$, define the unexplained separation $c = (2 - \kappa\sqrt{2}a)_+$. Proposition 3 then gives

$$R^* = p(1 - p)c^2.$$

Here R^* denotes the minimum MSE over the benchmark's constrained predictors. The two target values, representation family, and support are specified in advance by the benchmark design.

An independent training sample gives the empirical frequency $\widehat{p}$ of state 1. We choose the constrained empirical-risk minimizer whose predictions at the two states are

$$\widehat{f}(0) = -1 + \widehat{p}c, \qquad \widehat{f}(1) = 1 - (1 - \widehat{p})c. \tag{32}$$

These values lie in $[-1, 1]$ and have separation $2 - c = \min\{2, \kappa\sqrt{2}a\}$. The affine construction in Proposition 3 therefore realizes them within the budget; when $a = 0$, the two values coincide. If one state is absent from training, (32) selects one feasible empirical minimizer using the predeclared support. This convention specifies the prediction even at an unobserved state.

Conditional on the training sample, state 0 has prediction error $-\widehat{p}c$, and state 1 has error $(1-\widehat{p})c$. Consequently the exact fitted population risk is

$$R(\widehat{f}) = c^2\{(1 - p)\widehat{p}^2 + p(1 - \widehat{p})^2\} = R^* + c^2(\widehat{p} - p)^2. \tag{33}$$

For an independent pair of calibration states, the witness score equals $c^2/2$ when the states differ and zero otherwise. The mismatch probability is $2p(1-p)$, so the expected score is exactly R^*. With n calibration observations, the pair lower bound uses $\lfloor n/2 \rfloor$ disjoint pairs. An odd final observation is excluded from pairing but remains in the fitted-risk and centered statistics. The reported calibration sizes are all even.

The centered comparator. For comparison, apply a centered certificate based on the target and representation variances, with Euclidean distance and a single reference class. Let $Z_1, \ldots, Z_n$ be independent calibration states, with $n \geq 4$, and $\bar{z} = n^{-1} \sum_{i=1}^{n} Z_i$ their empirical frequency. The output and representation variances, computed with denominator n, are

$$\widehat{s} = 4\bar{z}(1-\bar{z}), \qquad \widehat{t} = 2a^2\bar{z}(1-\bar{z}).$$

Their population counterparts are $\sigma^2 = 4p(1-p)$ and $\tau^2 = 2a^2p(1-p)$. The population centered bound is $(\sigma - \kappa\tau)_+^2 \leq R^*$, using nonnegative square roots. In this two-state benchmark the inequality is an equality.

For a failure probability $\eta \in (0,1)$, define

$$\begin{aligned} r &= \log(2/\eta)/n, \\ V_\eta &= (\widehat{s} - 4\sqrt{r/2})_+, \\ T_\eta &= \min\{1, \widehat{t} + \sqrt{2r} + 1/n\}, \\ L_\eta^{\mathrm{ctr}} &= (\sqrt{V_\eta} - \kappa\sqrt{T_\eta})_+^2. \end{aligned} \tag{34}$$

Here V_η is a lower bound on σ^2, T_η is an upper bound on τ^2, and the superscript "ctr" identifies the centered risk bound. We show that $L_\eta^{\mathrm{ctr}} \leq R^*$ with probability at least $1-\eta$.

First, $\mathbb{E}\widehat{s} = (1-1/n)\sigma^2 \leq \sigma^2$, and changing one calibration observation changes $\widehat{s}$ by at most $4/n$. The bounded-differences inequality therefore gives

$$\widehat{s} - \mathbb{E}\widehat{s} \leq 4\sqrt{r/2}$$

with probability at least $1 - \eta/2$. On this confidence event, $V_\eta \leq \sigma^2$. Second, changing one observation changes $\widehat{t}$ by at most $2/n$, so the same inequality gives

$$\mathbb{E}\widehat{t} - \widehat{t} \leq \sqrt{2r}$$

with failure probability at most $\eta/2$. Since $\mathbb{E}\widehat{t} = (1-1/n)\tau^2$ and $\tau^2 \leq 1$,

$$\tau^2 \leq \widehat{t} + \sqrt{2r} + \tau^2/n \leq \widehat{t} + \sqrt{2r} + 1/n.$$

Capping the right side at one remains valid because $\tau^2 \leq 1$. A union bound gives $V_\eta \leq \sigma^2$ and $T_\eta \geq \tau^2$ jointly with probability at least $1-\eta$. The expression in (34) increases with its first variance argument and decreases with its second. Hence it is no larger than $(\sigma - \kappa\tau)_+^2 \leq R^*$ on that event. The constant one is the general simplex-variance bound; it is conservative for this two-coordinate example.

In each Monte Carlo trial, the fitted-risk upper bound, pair-witness lower bound, and centered lower bound each receive failure probability $\delta/3$. For the centered bound, setting $\eta = \delta/3$ splits that allocation between its two internal variance bounds. The three reported directions therefore hold jointly with probability at least $1-\delta$ within a trial. No joint $1-\delta$ event is asserted across all trials, grid conditions, or separately reported experiments.

References

David Abel, David Hershkowitz, and Michael Littman. Near optimal behavior via approximate state abstraction. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 2915–2923, 2016. URL https://proceedings.mlr.press/v48/abel16.html.

Aditya Kumar Akash, Vishnu Suresh Lokhande, Sathya N. Ravi, and Vikas Singh. Learning invariant representations using inverse contrastive loss. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 6582–6591, 2021. doi: 10.1609/aaai.v35i8.16815.

E. Alpaydin and Fevzi Alimoglu. Pen-based recognition of handwritten digits. UCI Machine Learning Repository, 1996. URL https://archive.ics.uci.edu/dataset/81/pen+based+recognition+of+handwritten+digits.

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *The Annals of Applied Statistics*, 19(2):1641–1662, 2025. doi: 10.1214/24-AOAS1998. URL https://arxiv.org/abs/2110.01052.

Francis R. Bach. Consistency of the group lasso and multiple kernel learning. *Journal of Machine Learning Research*, 9(40):1179–1225, 2008. URL https://jmlr.org/papers/v9/bach08b.html.

Avrim Blum and Anh Tuan Nguyen. Robustly reliable learning for regression. arXiv:2610.17381v2, 2026. URL https://arxiv.org/abs/2610.17381v2.

Peter Bühlmann and Bin Yu. Boosting with the L_2 loss: Regression and classification. *Journal of the American Statistical Association*, 98(462):324–339, 2003. doi: 10.1198/016214503000125. URL https://stat.ethz.ch/Manuscripts/buhlmann/boosting.rev5.pdf.

Gustav Conradie, Nicolas Boullé, Jean-Christophe Loiseau, Steven L. Brunton, and Matthew J. Colbrook. Trustworthy Koopman operator learning: Invariance diagnostics and error bounds. arXiv:2603.15091, 2026. URL https://arxiv.org/abs/2603.15091.

George Coote and Matthew J. Colbrook. Residual-guided dictionary learning for spectrally accurate Koopman approximation. arXiv:2606.29083, 2026. URL https://arxiv.org/abs/2606.29083.

Dafydd Evans and Antonia J. Jones. A proof of the Gamma test. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 458(2027):2759–2799, 2002. doi: 10.1098/rspa.2002.1010. URL https://doi.org/10.1098/rspa.2002.1010.

Carles Gelada, Saurabh Kumar, Jacob Buckman, Ofir Nachum, and Marc G. Bellemare. DeepMDP: Learning continuous latent space models for representation learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2170–2179, 2019. URL https://proceedings.mlr.press/v97/gelada19a.html.

Lee-Ad Gottlieb, Aryeh Kontorovich, and Robert Krauthgamer. Efficient regression in metric spaces via approximate lipschitz extension. *IEEE Transactions on Information Theory*, 63(8):4838–4849, 2017. doi: 10.1109/TIT.2017.2713820.

Masih Haseli and Jorge Cortés. Temporal forward–backward consistency, not residual error, measures the prediction accuracy of extended dynamic mode decomposition. *IEEE Control Systems Letters*, 7:649–654, 2023. doi: 10.1109/LCSYS.2022.3214476. URL https://par.nsf.gov/servlets/purl/10466905.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830. URL https://www.tandfonline.com/doi/abs/10.1080/01621459.1963.10500830.

Zeyuan Jin, Mohammad Khajenejad, and Sze Zheng Yong. Data-driven model invalidation for unknown Lipschitz continuous systems via abstraction. In *2020 American Control Conference*, pages 2975–2980, 2020. doi: 10.23919/ACC45564.2020.9147725. URL https://arxiv.org/abs/2003.11662.

Michael I. Jordan and Robert A. Jacobs. Hierarchical mixtures of experts and the EM algorithm. *Neural Computation*, 6(2):181–214, 1994. doi: 10.1162/neco.1994.6.2.181. URL https://doi.org/10.1162/neco.1994.6.2.181.

Vladimir Kostic, Pietro Novelli, Andreas Maurer, Carlo Ciliberto, Lorenzo Rosasco, and Massimiliano Pontil. Learning dynamical systems via Koopman operator regression in reproducing kernel hilbert spaces. In *Advances in Neural Information Processing Systems*, volume 35, pages 4017–4031, 2022. URL https://papers.nips.cc/paper_files/paper/2022/hash/196c4e02b7464c554f0f5646af5d502e-Abstract-Conference.html.

Mohit Kumar and Somayeh Kargaran. Prediction limits and Koopman closure of geometry-induced soft state abstractions. arXiv:2609.32652v3, 2026. URL https://arxiv.org/abs/2609.32652v3.

Mohit Kumar, Bernhard A. Moser, and Lukas Fischer. On mitigating the utility-loss in differentially private learning: A new perspective by a geometrically inspired kernel approach. *Journal of Artificial Intelligence Research*, 79:515–567, 2024. doi: 10.1613/jair.1.15071. URL https://jair.org/index.php/jair/article/view/15071.

Mohit Kumar, Alexander Valentinitsch, Magdalena Fuchs, Mathias Brucker, Juliana Bowles, Adnan Husakovic, Ali Abbas, and Bernhard A. Moser. Geometrically inspired kernel machines for collaborative learning beyond gradient descent. *Journal of Artificial Intelligence Research*, 83 (16), 2025. doi: 10.1613/jair.1.16821. URL https://jair.org/index.php/jair/article/view/16821.

Qianxiao Li, Felix Dietrich, Erik M. Bollt, and Ioannis G. Kevrekidis. Extended dynamic mode decomposition with dictionary learning: A data-driven adaptive spectral decomposition of the Koopman operator. *Chaos*, 27(10):103111, 2017. doi: 10.1063/1.4993854. URL https://arxiv.org/abs/1707.00225.

Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory*, 2009. URL https://arxiv.org/abs/0907.3740.

Daniel J. McDonald, Cosma Rohilla Shalizi, and Mark Schervish. Nonparametric risk bounds for time-series forecasting. *Journal of Machine Learning Research*, 18(32):1–40, 2017. URL https://www.jmlr.org/papers/v18/13-336.html.

Mario Milanese and Carlo Novara. Set membership identification of nonlinear systems. *Automatica*, 40(6):957–975, 2004. doi: 10.1016/j.automatica.2004.02.002. URL https://www.sciencedirect.com/science/article/pii/S0005109804000470.

Manuel M. Müller, Yuetian Luo, and Rina Foygel Barber. Are all models wrong? fundamental limits in distribution-free empirical model falsification. In *Proceedings of the 38th Conference on Learning Theory*, volume 291 of *Proceedings of Machine Learning Research*, pages 4271–4308, 2025. URL https://proceedings.mlr.press/v291/muller25a.html.

Roberto I. Oliveira, Paulo Orenstein, Thiago Ramos, and João Vitor Romano. Split conformal prediction and non-exchangeable data. *Journal of Machine Learning Research*, 25(225):1–38, 2024. URL https://www.jmlr.org/papers/v25/23-1553.html.

Satinder P. Singh, Tommi Jaakkola, and Michael I. Jordan. Reinforcement learning with soft state aggregation. In *Advances in Neural Information Processing Systems*, volume 7, pages 361–368, 1994. URL https://papers.neurips.cc/paper_files/paper/1994/hash/287e03db1d99e0ec2edb90d079e142f3-Abstract.html.

TensorFlow. Simple audio recognition: Recognizing keywords, 2026. URL https://www.tensorflow.org/tutorials/audio/simple_audio. Official Mini Speech Commands subset and tutorial. Accessed October 29, 2026.

Pete Warden. Speech commands: A dataset for limited-vocabulary speech recognition. *arXiv preprint arXiv:1804.03209*, 2018. URL https://arxiv.org/abs/1804.03209.

Matthew O. Williams, Ioannis G. Kevrekidis, and Clarence W. Rowley. A data-driven approximation of the Koopman operator: Extending dynamic mode decomposition. *Journal of Nonlinear Science*, 25(6):1307–1346, 2015. doi: 10.1007/s00332-015-9258-5. URL https://arxiv.org/abs/1408.4408.

Liming Wu and Sen Yang. Non-asymptotic estimates of the minimal risk in statistical learning. arXiv:2606.23295, 2026. URL https://arxiv.org/abs/2606.23295.